Point Cloud Retrieval with PQ-Chamfer Distance and Graph Regularization: Training-Free Geometric Operators with a Supervised Per-Query Fusion-Weight Predictor for Weak-Baseline LLM Embeddings

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Abstract

General-purpose large language model (LLM) hidden states, when used directly for retrieval via mean-pooled centroid cosine, give baselines weaker than retrieval-trained dense encoders. We ask whether *geometric post-processing*—training-free distance and smoothing operators composed with a supervised per-query fusion-weight predictor—can close part of this gap on weak-baseline LLM embeddings.

We introduce a two-stage geometric reranking pipeline on token-level point clouds extracted from a frozen Qwen3-8B-Q4. Stage 1, **PQ-Chamfer distance**, exploits the product-quantization decomposition $\mathbb{R}^{4096} = \bigoplus_{s=1}^{64} \mathbb{R}^{64}$, computes exact per-subspace cosine distances, and averages them, breaking concentration of measure without quantization error. Stage 2, **graph Laplacian smoothing** on a k-NN document graph built from PQ-Chamfer distances, propagates relevance across inter-document structure. The per-query fusion weight is learned by a supervised gradient-boosted regressor (GBM-11) under a 50/50 train/test split.

On five non-Quora BEIR corpora (NFCorpus, ArguAna, SciFact, SCIDOCS, FiQA), the pipeline lifts the Qwen3-8B token-centroid cosine baseline by +20.9% to +157.4% NDCG@10. Paired bootstrap against the per-corpus oracle reference gives $p < 0.001$ on four corpora and $p < 0.005$ on SciFact (Cohen's d ranges 0.20–0.72). Against retrieval-trained baselines on identical splits in our own infrastructure (audited rebuild after a vector-provenance fix), the pipeline is competitive on weak-baseline corpora but does not displace cross-encoders on short scientific abstracts. This positions the contribution as *geometric post-processing on weak-baseline LLM embeddings*, a regime-specific com-

plement. This work additionally documents 21 falsified approaches (11 with complete raw experimental logs preserved from the development of this research program; 10 retrospectively reconstructed from contemporaneous notes with provenance flagged per-entry) organized into a three-class failure-mechanism taxonomy. The full pipeline cost $47 on a single consumer GPU.

Keywords: dense retrieval, LLM embeddings, product quantization, graph Laplacian smoothing, BEIR benchmark

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1. Introduction

Retrieval-Augmented Generation (RAG) has become the dominant paradigm for grounding large language models in external knowledge [19, 13]. The standard RAG pipeline employs a bi-encoder to map queries and documents into a shared embedding space, then ranks candidates by cosine similarity. Let $\mathbf{v}_q \in \mathbb{R}^d$ and $\mathbf{v}_i \in \mathbb{R}^d$ denote the query and document embeddings, respectively:

$$\text{score}(d_i) = \frac{\mathbf{v}_q \cdot \mathbf{v}_i}{\|\mathbf{v}_q\| \, \|\mathbf{v}_i\|} \tag{1}$$

This formulation has two fundamental limitations. First, it represents each document as a *single point* in $\mathbb{R}^d$, conflating diverse semantic facets into one centroid. Second, it scores each document independently, ignoring inter-document structure—topic clusters, semantic neighborhoods, and distributional patterns—that the retrieved set collectively encodes.

Existing training-free remedies do not address these two limitations directly. Query-rewriting methods such as HyDE [9] and Query2Doc [39] alter the query but leave the document representation intact, so cosine concentration in $\mathbb{R}^d$ persists at scoring time. Sparse–dense fusion with BM25 [31] adds lexical signal but does not recover sub-document structure, and shares cosine's pointwise scoring assumption on the dense side. Pseudo-relevance feedback (PRF) over dense embeddings [20] expands the query in embedding space, again inheriting concentration of measure. Late-interaction architectures such as ColBERT and ColBERTv2 [17, 34] explicitly preserve token-level structure

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but require dedicated training of a token-aware encoder, contradicting the operator-level training-free regime we target. The gap we identify, therefore, is the absence of a method that (a) exploits the token-level geometry of frozen general-purpose embeddings and (b) scores documents jointly through inter-document structure—without any retrieval-specific training.

This gap is addressed by treating documents as point clouds rather than points [2]. The per-token hidden states of a language model form a cloud in embedding space whose spread, density variation, and subspace projections carry semantic information that a single centroid discards. Two design questions follow: what distance metric is appropriate between document point clouds, and how can inter-document structure be incorporated without the cost of cross-encoder inference? Both are answered with training-free geometric operations: PQ-Chamfer distance for the first, graph Laplacian smoothing for the second.

Relation to prior versions. This paper extends the Shape-CFD research program [2, 3]. Earlier iterations of this research program explored convection-diffusion equations and sentence-level point clouds. The present work abandons the convection-diffusion narrative—experimentally falsified across five independent validations (Section 4.12)—and reorganizes the framework around token-level point clouds, PQ-Chamfer distance, and graph Laplacian smoothing.

1.1. Contributions

This work positions itself as a *geometric post-processing layer* along a **baseline-strength gradient**: across the spectrum from weak frozen-LLM cosine baselines (Qwen3-8B-Q4 token-mean centroid, NDCG@10 $\sim$ 0.17–0.45 across 5 BEIR corpora), through retrieval-trained dense baselines (BGE-M3, BGE-large-en-v1.5, NDCG@10 $\sim$ 0.31–0.64), to retrieval-trained late-interaction baselines (ColBERTv2 MaxSim, NDCG@10 $\sim$ 0.31–0.61), the magnitude of ΔNDCG@10 from geometric post-processing varies non-monotonically with both baseline regime and graph topology quality (e.g., on the same-encoder Qwen3-8B-Q4 main results in Table 3, ArguAna's argumentative-text query/document pairs yield rich k-NN graph topology and the largest $|\Delta|$ in absolute NDCG@10 terms despite a mid-range cosine baseline, while SCIDOCS' sparse short cite-prediction text yields noisier graph topology and a smaller absolute $|\Delta|$ than its

weak baseline alone would predict; per-corpus magnitudes depend jointly on baseline-regime headroom and corpus-level graph-topology signal-to-noise), while the sign of Δ depends on the baseline regime (uniformly positive on weak baselines; mixed on strong baselines, with the third-tier ColBERTv2 + Lap result reported in Section 4.14). This baseline-strength gradient—rather than a universal "always-positive" claim—is the central evidence pattern of our work, and it positions the geometric post-processor as a *boundary-positioned* tool that *captures retrieval signal complementary to (not orthogonal to) trained dense and late-interaction priors*, with the boundary being most pronounced on weak frozen-LLM baselines.

Framing of contribution. This paper is positioned as an *engineering and boundary-characterisation study* rather than a conceptual-breakthrough method paper. No single ingredient (the subspace decomposition, the symmetric averaged-cosine aggregator, graph Laplacian smoothing on a k-NN graph, or the supervised fusion-weight regressor) is, taken in isolation, a methodological novelty: product quantization for distance computation has prior literature [15]; symmetric token-level aggregators appear in SBERT-family token similarity, WMD-style optimal transport, and BERTScore; graph Laplacian smoothing on retrieval point-cloud graphs has been used in graph-augmented and graph-of-thought retrieval; supervised per-query fusion weight prediction is a standard learning-to-rank formulation (e.g., LambdaMART). The paper's contribution is the *boundary characterization* (a cosine-strength gradient and a graph-topology dependence that jointly govern when the geometric pipeline helps and when it does not) plus the *systematic falsification taxonomy* (21 falsified approaches—11 with complete raw experimental logs and 10 retrospectively reconstructed with per-entry provenance flagging—organised into three failure-mechanism classes) that surface design constraints not articulated in prior work for the frozen-LLM-token-cloud retrieval regime. Our contributions are:

1. **Niche characterization: geometric post-processing for weak-baseline LLM embeddings.** We characterise a specific application niche: when the first-stage cosine baseline is weak (NDCG@10 < 0.3, typical of frozen general-purpose LLM token-mean centroids), a geometric pipeline whose distance and smoothing operators are training-free and whose per-query fusion weight is supervised by a gradient-boosted (GBM-11)

regressor (Section 4.8) lifts retrieval quality by +20.9% (SciFact, against sentence-level cosine 0.4483; the SciFact gain against the token-level cosine reference 0.4701 used internally in Table 4 is +15.3%) to +157.4% (FiQA) over the same-encoder cosine baseline. Compared on identical splits to retrieval-trained baselines (BGE-M3, BGE-large-en-v1.5, BGE-reranker-v2-m3, ColBERTv2; audited rebuild after a vector-provenance fix), under the strictly protocol-matched single-split lens (both Ours and baselines on identical single-split single-seed full evaluation; see Section 4.4) the pipeline ranks *first* on SCIDOCS (0.2147 vs. next-best BGE-reranker-v2-m3 0.1823, +17.8%), *second* on NFCorpus (0.3270 vs. 0.3275, −0.15%), ArguAna (0.4418 vs. 0.4993, −11.5%), and FiQA (0.3977 vs. 0.4227, −5.9%), and *fourth* on SciFact (0.4946, behind ColBERTv2 / BGE-M3 / BGE-reranker, where retrieval-trained cross-encoders dominate on short scientific abstracts). As a robustness cross-check, under our deployment-time 5-seed 50/50 protocol (a less-noisy 5-seed-mean point estimate, but *not* strictly protocol-matched against the single-split single-seed audited baselines) the pipeline ranks first on three of five non-Quora corpora—NFCorpus (0.3297 vs. 0.3275, +0.7%), SCIDOCS (0.2451 vs. 0.1823, +34.4%), and FiQA (0.4331 vs. 0.4227, +2.5%)—second on ArguAna (0.4862 vs. 0.4993, −2.6%), and third on SciFact (0.5418 vs. 0.7228, −25.0%). Both lenses are reported transparently to forestall a protocol-mismatch concern, with Lens 2 (protocol-matched) as the primary headline and Lens 1 (5-seed deployment) as the robustness cross-check. The contribution is the systematic mapping of *when geometric post-processing helps and when it does not*, with SciFact's residual gap and token-level PQ-Chamfer's −3.1% on the strong-cosine SciFact baseline (Section 5.3) serving as direct evidence for the weak-baseline boundary. **Scope of the "training-free" label.** The geometric operators (PQ-Chamfer distance and graph Laplacian smoothing) are training-free; the per-query fusion weight λ is learned by a supervised gradient-boosted regressor (GBM-11) on per-query oracle $\lambda^\star$ targets computed from `qrels` under a 50/50 train/test split (Section 4.8). Deployment-time alternatives that do not require oracle $\lambda^\star$ are explored as a LinUCB online policy on the single NFCorpus corpus as a proof-of-concept; full deployment-time online learning is identified as future work.

2. **Geometric post-processing pipeline (two-stage core on frozen-LLM token**

point clouds, plus an optional centroid coarse filter). Two core geometric operations are composed on per-token hidden states from a frozen Qwen3-8B (`-pooling none`; document point clouds with median ∼356 tokens, no retrieval-specific training) and characterize their behaviour as a niche post-processing layer rather than as a method applicable to all encoder regimes: (a) *PQ-Chamfer distance*—we use product quantization [15] from a compression primitive to a distance primitive[1], decomposing $\mathbb{R}^{4096} = \bigoplus_{s=1}^{64} \mathbb{R}^{64}$, computing exact subspace cosine distances and averaging across $M = 64$ subspaces; this breaks concentration of measure with no quantization error (measured $\sigma_{\text{sub}}/\sigma_{\text{full}}$ ratios 2.0×/1.61×/1.54× for Qwen3-8B/BGE-M3/BGE-large, audited rebuild; Section 3.2); geometrically, MaxSim "votes for the best per-token match" (asymmetric, full-dimensional) while PQ-Chamfer "recognizes the overall geometric shape" via subspace-decomposed averaged distance (symmetric, subspace-projected); (b) *token-level point cloud retrieval on frozen LLM hidden states*—PQ-Chamfer is applied directly to per-token hidden states without retrieval-specific fine-tuning, distinguishing the operator from ColBERT/ColBERTv2 [17, 34] which require a trained token-aware encoder; (c) *graph Laplacian smoothing* on a k-NN document graph built from PQ-Chamfer distances (not centroid cosine), iterated as $C^{(t+1)} = (I - \alpha L)C^{(t)}$. On the same-encoder Qwen3-8B configuration, graph Laplacian smoothing yields +5.8% to +44.2% NDCG@10 on all six BEIR datasets evaluated (NFCorpus, ArguAna, SciFact, SCIDOCS, FiQA, and Quora) relative to the same-encoder cosine baseline; the five non-Quora corpora carry the GBM-11 deployment-rule headline because token-level Qwen3-8B point cloud data is fully available on those corpora. Cross-encoder applicability is more nuanced: under cross-model per-(corpus, model) tuning (Section 4.13), smoothing helps on 1 of 5 corpus–model pairs in our heterogeneous-encoder grid and degrades on the others; the smoother is therefore characterised as same-encoder consistent and cross-encoder tuning-required—a regime-specific tool requiring per-(model, corpus) tuning.

3. **Supervised per-query fusion weight on top of the training-free geometric operators.** The manually-set fusion weight is replaced λ with a gradient-boosted

[1] Chamfer distance is a semimetric (Lemma 1); we use "distance" following point cloud convention.

regressor (GBM, 11 features) that predicts λ from cheap per-query proxies. Macro NDCG@10 across the five BEIR corpora rises from 0.3653 (fixed $\lambda = 0.3$) to 0.4072 (+11.5% relative-to-fixed; Section 4.8), closing 69% of the oracle–fixed gap. The training data are per-query oracle $\lambda^\star$ targets computed from `qrels` under a 50/50 train/test split. The remaining oracle gap motivates an online incremental-SGD extension that is sketched in Section 4.8 and report a proof-of-concept LinUCB replay run for on NFCorpus (single-corpus PoC); full deployment-time online learning is identified as future work. We also explore a forward, training-free Hartree-style self-consistent variant (**2 free parameters** versus **11** for GBM-11; Section 4.8, Eqn. (14)) that determines per-query $\lambda^{\text{eff}}(q)$ as a fixed point of the retrieval-norm closure without requiring oracle $\lambda^\star$ supervision; this variant matches fixed $\lambda = 0.3$ on NFCorpus (0.3106 vs 0.3116) but underperforms GBM-11 (0.3297) by 5.8% relative, reported transparently as a negative result that establishes the $11 \to 2$ free-parameter reduction is achievable in form but not at parity retrieval quality. The supervised fusion weight is the *only* learned component of the pipeline; the distance and smoothing operators remain training-free.

4. **Systematic falsification with a three-class failure-mechanism taxonomy.** We document 21 falsified approaches—**11 with complete raw experimental logs preserved from the development of this research program (per-query NDCG@10, configuration, notes); 10 retrospectively reconstructed from contemporaneous notes with provenance flagged per-entry** (Supplementary Material A.3). The 21 entries are organized into three failure-mechanism classes—curse-of-orthogonality / random-projection failures; saturation-on-strong-baseline failures; signal-cancellation / information-redundancy failures—with mathematical mechanism analysis per class. Together with the boundary characterization (item 1) this falsification taxonomy is the paper's main substantive contribution: it surfaces design constraints (the cosine-strength gradient, the graph-topology dependence, the saturation-on-strong-baseline barrier) that any geometric post-processor on frozen-LLM token clouds must respect. Audit-grade reproducibility—including the independent vector-provenance audit that revealed 19.93%/18.62% of NFCorpus/SciFact corpus documents in earlier BGE-M3 vector exports were silently zero-encoded (root cause and Pearson $r \approx 0.998$ propagation analysis in Section 4.13 *Vector-provenance*

note; all baseline numbers in Table 6 are the audited rebuild)—is reported as a methodological commitment in Section 4.13 rather than as a stand-alone contribution slot. Code, indices, and the full falsification log are released for anonymous review at `https://anonymous.4open.science/r/SHAPE-CFD-IS2026/`.

1.2. Research Objectives

This study addresses the following research questions:

1. **RQ1:** Can training-free geometric operations (PQ-Chamfer distance and graph Laplacian smoothing), composed with a supervised per-query fusion-weight predictor, on general-purpose LLM hidden states improve dense retrieval quality?

2. **RQ2:** How does the improvement transfer across different frozen embedding architectures (Qwen3-8B-Q4, BGE-M3, BGE-large-en-v1.5), and what are the boundaries of cross-encoder applicability under per-(corpus, model) tuning?

3. **RQ3:** At what computational cost can these improvements be achieved compared to training specialized retrieval models?

2. Literature Review

The work is positioned within five intersecting strands of prior research, then explicitly delineate the regime in which our specific design choices are, to our knowledge, new.

2.1. Multi-Vector and Point-Cloud Document Representations

The idea of representing documents as point clouds—collections of token-level or phrase-level vectors rather than a single centroid—is not new in information retrieval. Prior work is organised along the dimension that most distinguishes it from ours: whether the representation is *trained* or *frozen*.

Trained multi-vector retrieval. The dominant family is built around contrastive fine-tuning of token-aware encoders. ColBERT [17] introduced late interaction with the asymmetric MaxSim operator $\mathrm{MaxSim}(q, d) = \sum_{i \in q} \max_{j \in d} \langle q_i, d_j \rangle$ on top of a BERT encoder fine-tuned on MSMARCO query–document pairs. ColBERTv2 [34] compressed token embeddings via residual quantization with learned codebooks, and PLAID [33]

accelerated search through centroid-based candidate pruning. COIL [8] restricted late interaction to lexically-overlapping tokens, reducing token-token comparisons. ME-BERT [25] pooled multiple cluster centroids per document instead of a single CLS vector. DESSERT [5] accelerated multi-vector retrieval via locality-sensitive hashing on token vectors. All of these methods share two properties that distinguish them from our work: (i) the token-level encoder is contrastively fine-tuned on retrieval supervision, and (ii) the retrieval-time scoring uses an asymmetric or dot-product formulation rather than a symmetric subspace-averaged cosine.

Frozen multi-vector representations. A smaller line of work explores point-cloud retrieval on frozen encoders. Sentence-BERT [29] demonstrated that frozen pretrained encoders, when pooled appropriately, achieve competitive retrieval; however, the pooling collapses the document to a single vector and is not a point cloud. WMD [18] computed earth mover's distance between word-embedding clouds of two documents, but on word2vec rather than transformer hidden states and without graph regularization. BERTScore [41] applied one-directional max-pooling between two BERT-encoded sentences for evaluation rather than retrieval ranking. To our knowledge, no prior work computes (a) symmetric subspace-decomposed Chamfer distance on (b) frozen general-purpose LLM hidden states for (c) document retrieval ranking with (d) an additional inter-document graph regularization step. Our contribution is the specific combination, not any single ingredient.

Recent LLM-based embeddings. E5-Mistral [38] and GTE [23] fine-tune large LLMs on retrieval objectives to produce strong single-vector representations, but they collapse the per-token cloud into one centroid via mean pooling and still require contrastive training. Any such encoder is treated as a frozen feature extractor and apply geometric reranking on top of it; results across Qwen3-8B-Q4 (4096-d, frozen), BGE-large-en-v1.5 (1024-d, contrastive-trained, frozen at our use point), and BGE-M3 (1024-d, contrastive-trained, frozen) are reported in Section 4.13.

2.2. Late-Interaction Comparison Axes

Three structural axes pin down on which our pipeline differs from the ColBERT family:

(i) Representation source. Frozen general-purpose LLM hidden states are used (Qwen3, BGE, GTE) with no retrieval-specific contrastive training; ColBERT-family encoders are explicitly trained on query–document supervision.

(ii) Aggregation symmetry and metric structure. PQ-Chamfer is symmetric and operates on *subspace cosine distances* averaged across $M = 64$ subspaces:

$$d_{\mathrm{PQ}}(q, d) = \frac{1}{M} \sum_{s=1}^{M} \delta_s\big(q^{(s)}, d^{(s)}\big),$$

with δ_s the cosine distance in subspace s (Section 3). MaxSim is asymmetric, dot-product based, and operates on full-dimensional vectors. The subspace average breaks concentration of measure: the isotropic-Gaussian textbook prediction is a 7.8× discrimination gain (from ∼0.016 in $\mathbb{R}^{4096}$ to ∼0.125 per subspace), but the empirically measured $\sigma_{\mathrm{sub}}/\sigma_{\mathrm{full}}$ ratio on contextual LLM embeddings is 2.0× on Qwen3-8B / 1.61× on BGE-M3 / 1.54× on BGE-large (anisotropy depresses the gain; see Section 3.2). MaxSim does not exploit either subspace structure.

(iii) Inter-document signal. A graph Laplacian smoothing step is introduced that no late-interaction method performs: ColBERT-family scoring is per-pair (independent of other documents), whereas our pipeline additionally propagates relevance across a document KNN graph built from PQ-Chamfer distances.

2.3. Two Structural Novelties

Two design choices are, to our knowledge, novel in this combination, and they jointly anchor our positioning:

(N1) PQ *without* quantization, as an exact distance. Product quantization [15] is canonically a *compression* primitive: subspace decomposition $\mathbb{R}^D = \bigoplus_s \mathbb{R}^{D/M}$ is paired with per-subspace K-means codebooks to store quantized codes for approximate-nearest-neighbor search. ColBERTv2/PLAID inherit this trade-off (smaller storage, quantization-error penalty). The trade-off is inverted: the subspace decomposition is kept and *drop* the codebook, computing exact cosine distance within each subspace and averaging across subspaces. The benefit is variance inflation without quantization error; the cost is full-precision token storage. Quantization-as-storage and our distance-as-decomposition are complementary rather than competing; an integrated PQ-codebook variant of our pipeline is identified as a future-work direction (Section 6.4).

(N2) Graph Laplacian on a point-cloud-distance KNN graph. Standard graph reranking methods such as G-RAG [11] or GAR [26] build their KNN graphs from *single-vector similarities* (centroid cosine or BM25). The graph is built from *point-cloud distances* (PQ-Chamfer between document token clouds), giving the smoothing operator a graph topology that respects sub-document structure. To our knowledge, no prior graph-reranking method uses a token-level point-cloud distance to define graph edges.

LLM-based reranking. RankGPT [36] demonstrates that LLMs can serve as effective listwise rerankers via prompt engineering. RankLLM [28] provides a scalable framework for LLM-based reranking. AFR-Rank [24] improves efficiency by filtering noise documents before LLM reranking. A recent survey [21] traces the evolution from heuristic reranking to LLM-based approaches. These methods achieve strong results but require expensive API calls ($0.01–0.10 per query); our pipeline—training-free geometric operators with a supervised per-query fusion-weight predictor (Section 4.8)—achieves competitive results at orders of magnitude lower cost in the weak-baseline LLM embedding niche.

Graph-based reranking. Graph methods span from PageRank [27] to diffusion-based RAG approaches [4]. G-RAG [11] learns graph neural network rerankers, and GAR [26] uses pseudo-relevance feedback on document graphs. Our Laplacian smoothing operates on point-cloud-derived distances and requires no training.

Product quantization. PQ [15] and systems built on it (FAISS [16], ScaNN [12], OPQ [10]) use subspace decomposition for compression and approximate search. The decomposition is repurposed for exact distance computation, using subspace structure to break concentration of measure.

Point cloud distances. Chamfer distance [7] and EMD [32] are the dominant 3D point cloud metrics. Chamfer corresponds to unbalanced optimal transport with fully relaxed marginals [35], making it effective for documents of different lengths. BERTScore [41] applies a similar one-directional matching; WMD [18] uses EMD over word embeddings.

3. Methodology

3.1. Document Point Cloud Representation

Let $\mathcal{M}$ denote a pretrained language model (Qwen3-8B, $d = 4096$). For a document d_i consisting of tokens $\{t_{i,1}, t_{i,2}, \ldots, t_{i,n_i}\}$, the last hidden states are extracted with pooling disabled (`-pooling none`):

$$\mathcal{P}_i = \{h_{i,1}, h_{i,2}, \ldots, h_{i,n_i}\} \subset \mathbb{R}^d \tag{2}$$

where $h_{i,j} \in \mathbb{R}^{4096}$ is the last-layer hidden state for token $t_{i,j}$. Similarly, a query q with tokens $\{t_{q,1}, \ldots, t_{q,m}\}$ yields a query point cloud $\mathcal{P}_q = \{h_{q,1}, \ldots, h_{q,m}\}$.

In our NFCorpus evaluation, the average document contains $\sim$356 tokens and the average query contains $\sim$6 tokens. The entire NFCorpus token cloud database (2,473 documents, 880,791 tokens) occupies 14 GB in SQLite storage.

Centroid representation. For coarse-grained operations, the centroid is also computed of each point cloud:

$$\bar{v}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} h_{i,j} \tag{3}$$

This centroid serves as the single-vector representation for initial retrieval and coarse filtering.

3.2. PQ-Chamfer Distance

PQ-Chamfer distance is defined in four stages, presented incrementally below: (i) subspace decomposition of the ambient embedding space; (ii) Chamfer aggregation across two point clouds; (iii) a boundedness lemma establishing the operator's metric structure; and (iv) a VT-Aligned variant that reverses the aggregation order. Each stage introduces one operator with two-to-three sentences of intuition before its formal definition.

Stage I. Subspace decomposition of the ambient embedding space. The geometric reason for splitting the 4096-d space is concentration-of-measure: in high ambient dimension cosines collapse toward a narrow band and lose discriminative resolution, whereas 64-d sub-cosines retain spread (measured in Section 3.2). Concretely, the 4096-dimensional embedding space is partitioned into $M = 64$ contiguous subspaces of

dimension $d_s = 64$ each. For a vector $h \in \mathbb{R}^{4096}$, we write $h^{(s)}$ for the projection onto the s-th subspace:

$$h^{(s)} = (h_{(s-1)d_s+1}, \ldots, h_{s \cdot d_s}) \in \mathbb{R}^{d_s}, \quad s = 1, \ldots, M \tag{4}$$

The cosine distance within the s-th subspace is denoted δ_s to distinguish it from the subspace dimension d_s:

$$\delta_s(a, b) = 1 - \frac{a^{(s)} \cdot b^{(s)}}{\|a^{(s)}\| \, \|b^{(s)}\|} \tag{5}$$

The PQ distance between two tokens is the average across subspaces:

$$d_{\mathrm{PQ}}(a, b) = \frac{1}{M} \sum_{s=1}^{M} \delta_s(a, b) \tag{6}$$

Why subspace decomposition works (measured, not assumed). The isotropic-Gaussian textbook prediction is $\sigma \approx 1/\sqrt{d} = 0.016$ for $d = 4096$ and $\sigma_s \approx 1/\sqrt{64} = 0.125$ for $d_s = 64$, giving a discrimination ratio of $\sqrt{M} = 8.0\times$. However, contextual LLM embeddings are well-documented to be *anisotropic* [6, 1], so the isotropic prediction does not directly apply.

Concentration-of-measure: empirical anisotropy measurement.. We characterise pairwise-cosine standard deviation on Qwen3-8B sentence-level centroid embeddings from NFCorpus, using 2,000 documents and 10,000 random pairs per measurement (`benchmark/measure_sigma_subspace.py`). The full 4096-d cosine has mean $= 0.76$, indicating substantial anisotropy (an isotropic Gaussian gives mean near zero), and $\sigma_{\mathrm{full}} = 0.058$. The per-subspace 64-d cosine, averaged across all 64 contiguous subspaces, gives $\sigma_{\mathrm{sub}} = 0.118$. This matches the isotropic-subspace prediction of 0.125 within 5%, with per-subspace σ ranging $[0.027, 0.195]$. The measured discrimination ratio is $\sigma_{\mathrm{sub}}/\sigma_{\mathrm{full}} = 2.0\times$, lower than the isotropic prediction of $8.0\times$ but strictly positive. Subspace decomposition does break the global anisotropy in practice, just less dramatically than isotropic Gaussian theory suggests. Note that this is not a random projection (which proved harmful in our tests; see Section 4.12): the subspace boundaries respect the model's dimensional structure. Ablation confirms that contiguous 64×64 outperforms 32×128 and 16×256. Ordered subspaces perform comparably to randomly permuted ones. This indicates that Qwen3's embedding dimensions are $\approx$ uniform in

within-subspace information content, even though the global geometry is anisotropic. Full output: `benchmark/data/results/sigma_subspace_measurement.json`.

Cross-model anisotropy verification (audited rebuild). The same measurement was repeated on BGE-M3 (audited-rebuild vectors, see Section 4.13 *Vector-provenance note*) and BGE-large-en-v1.5 NFCorpus corpus embeddings (each 1024-d, partitioned into $M = 16$ contiguous 64-d subspaces). Results: BGE-M3 $\sigma_{\text{sub}}/\sigma_{\text{full}} =$ **1.61**$\times$ (full $\sigma = 0.068$, subspace mean $\sigma = 0.110$, audited rebuild post zero-vector fix); BGE-large $\sigma_{\text{sub}}/\sigma_{\text{full}} = 1.54\times$ (full $\sigma = 0.063$, subspace mean $\sigma = 0.097$). Both are below the corresponding isotropic prediction $\sqrt{16} = 4.0\times$ but strictly positive — confirming that the anisotropy plus subspace-discrimination pattern is *consistent across LLM-derived dense encoders* (Qwen3-8B 4096-d $\sigma_{\text{sub}}/\sigma_{\text{full}} = 2.0\times$, BGE-M3 NFCorpus $1.61\times$ [5-corpus mean $1.66\times$, range 1.57–$1.80\times$; see Table 1], BGE-large NFCorpus $1.54\times$), not a Qwen3-specific phenomenon. The pre-audit BGE-M3 ratio reported in earlier drafts ($2.70\times$) was inflated by 19.93% silently zero-encoded NFCorpus corpus documents in the prior BGE-M3 vectors, which depressed full-cosine σ relative to subspace σ; the audited rebuild gives the correct, more conservative value. Per-model output: `benchmark/data/results/sigma_bge_m3_v2_nfcorpus.json` (audited), `sigma_bge_large_nfcorpus.json`.

With the per-token PQ distance (6) in place, we now lift it to a distance between point clouds via Chamfer aggregation.

Stage II. Chamfer aggregation across point clouds. Given two point clouds $\mathcal{A}$ and $\mathcal{B}$, the PQ-Chamfer distance is:

$$d_{\text{PQC}}(\mathcal{A}, \mathcal{B}) = \frac{1}{2}\left[\frac{1}{|\mathcal{A}|}\sum_{a\in\mathcal{A}} \min_{b\in\mathcal{B}} d_{\text{PQ}}(a,b) + \frac{1}{|\mathcal{B}|}\sum_{b\in\mathcal{B}} \min_{a\in\mathcal{A}} d_{\text{PQ}}(a,b)\right] \tag{7}$$

The forward term $\frac{1}{|\mathcal{A}|}\sum_a \min_b$ measures how well $\mathcal{B}$ *covers* $\mathcal{A}$ (every point in $\mathcal{A}$ finds a nearby match in $\mathcal{B}$), and the backward term ensures symmetry.

Stage III. Boundedness and metric structure. Before Chamfer aggregation can be used inside a graph Laplacian smoothing operator (Section 3.3), we record the four metric properties that the smoothing argument actually requires.

Lemma 1 (PQ-Chamfer is a bounded symmetric semimetric, not a metric)**.** *For any non-empty point clouds $\mathcal{A}, \mathcal{B} \subseteq \mathbb{R}^d$ with $d = M \cdot d_s$, the PQ-Chamfer distance defined in*

Eq. (7) *satisfies:* (i) Boundedness: $d_{\mathrm{PQC}}(\mathcal{A},\mathcal{B}) \in [0,2]$*;* (ii) Symmetry: $d_{\mathrm{PQC}}(\mathcal{A},\mathcal{B}) = d_{\mathrm{PQC}}(\mathcal{B},\mathcal{A})$*;* (iii) Reflexivity: $d_{\mathrm{PQC}}(\mathcal{A},\mathcal{A}) = 0$*;* (iv) Triangle inequality does *not* hold in general.

Proof. (i) Each $\delta_s(a,b) = 1 - \cos\angle(a^{(s)}, b^{(s)}) \in [0,2]$, so $d_{\mathrm{PQ}}(a,b) = \frac{1}{M}\sum_s \delta_s \in [0,2]$. The Chamfer aggregation in Eq. (7) averages forward and backward min terms, each itself in $[0,2]$, giving $d_{\mathrm{PQC}} \in [0,2]$. (ii) The half-sum in Eq. (7) is symmetric by construction: swapping $\mathcal{A}$ and $\mathcal{B}$ swaps the two halves. (iii) For $a \in \mathcal{A}$, $\min_{b\in\mathcal{A}} d_{\mathrm{PQ}}(a,b) = d_{\mathrm{PQ}}(a,a) = 0$ (taking $b = a$); both halves of Eq. (7) reduce to zero. (iv) Counter-example: take $\mathcal{A} = \{e_1\}$, $\mathcal{B} = \{(e_1+e_2)/\sqrt{2}\}$, $\mathcal{C} = \{e_2\}$ in $\mathbb{R}^2$ (extended trivially to $\mathbb{R}^d$). Then $d_{\mathrm{PQC}}(\mathcal{A},\mathcal{B}) = d_{\mathrm{PQC}}(\mathcal{B},\mathcal{C}) = 1 - 1/\sqrt{2} \approx 0.293$, while $d_{\mathrm{PQC}}(\mathcal{A},\mathcal{C}) = 1 > 2\cdot 0.293 = 0.586$, violating $d_{\mathrm{PQC}}(\mathcal{A},\mathcal{C}) \le d_{\mathrm{PQC}}(\mathcal{A},\mathcal{B}) + d_{\mathrm{PQC}}(\mathcal{B},\mathcal{C})$. □

The KNN graph and Laplacian smoothing (Section 3.3) treat d_{PQC} as a *similarity proxy* rather than a true metric: the spectral argument used in Section 3.3 requires only that the symmetric edge weight $W_{ij} = \exp(-\beta d_{ij})$ be non-negative and bounded, both of which follow from properties (i)–(ii) of Lemma 1. Retrieval ranking requires only isotonicity of d_{PQC} with respect to relevance, not full metric structure (e.g., dot-product similarity, the de facto IR ranking signal, is also not a metric).

Stage IV. VT-Aligned variant (aggregation-order swap). Stages I–III define the operator used in our pipeline. We complete the four-stage map by recording one variant we ablated and dropped, so the design space surrounding PQ-Chamfer is explicit. The standard PQ-Chamfer computes subspace distances first and then applies the Chamfer min operator ("average-then-min"):

$$d_{\mathrm{PQC}}(\mathcal{A},\mathcal{B}) = \mathrm{Chamfer}\big(\mathcal{A},\mathcal{B};\ d_{\mathrm{PQ}}(a,b) = \tfrac{1}{M}\textstyle\sum_s \delta_s(a,b)\big) \tag{8}$$

VT-Aligned reverses the aggregation order ("min-then-average"): the minimization over the target set is performed *independently within each subspace*, allowing different subspaces to select different nearest neighbors:

$$d_{\mathrm{VT}}(\mathcal{A},\mathcal{B}) = \frac{1}{2}\left[\frac{1}{|\mathcal{A}|}\sum_{a\in\mathcal{A}}\frac{1}{M}\sum_{s=1}^{M}\min_{b\in\mathcal{B}}\delta_s(a,b) + \frac{1}{|\mathcal{B}|}\sum_{b\in\mathcal{B}}\frac{1}{M}\sum_{s=1}^{M}\min_{a\in\mathcal{A}}\delta_s(b,a)\right] \tag{9}$$

This captures finer-grained matching patterns than standard PQ-Chamfer, where a single globally nearest neighbor is used across all subspaces. Early ablation observed com-

parable performance for the VT-Aligned variant; the variant is omitted from the present pipeline due to lack of hypothesis-tested significance on the VT-Aligned vs. standard PQ-Chamfer contrast.

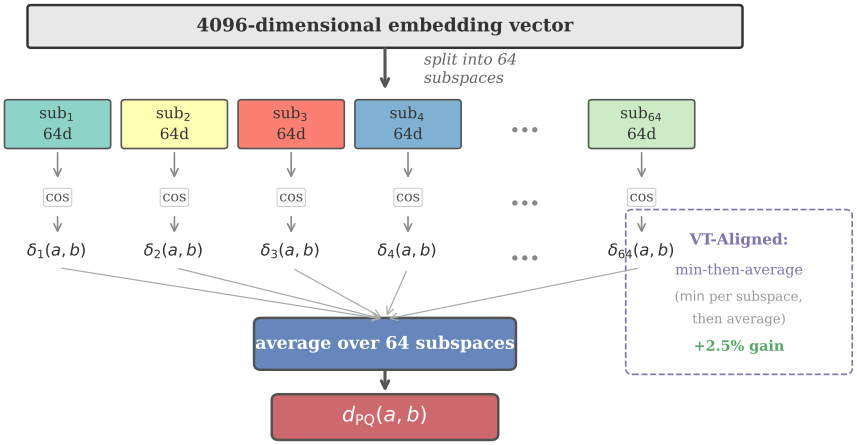

Figure 1: PQ-Chamfer subspace decomposition. A 4096-dimensional vector is partitioned into 64 contiguous 64-dimensional subspaces. Standard PQ-Chamfer averages subspace distances before applying the Chamfer min operator; VT-Aligned reverses this order; we omit this variant from the present pipeline (Section 3.2).

Connection to unbalanced optimal transport. Chamfer distance corresponds to UOT with fully relaxed marginals ($\tau \to 0$) [35]. This relaxation is essential for documents of different lengths: a 6-token query matched against a 356-token document need not force all document tokens to match, functioning as *hard semantic attention*.

3.3. Graph Laplacian Smoothing

Given a set of N candidate documents with initial relevance scores, a weighted KNN graph is constructed and iteratively smoothed.

Graph construction. A sparse undirected graph is built $\mathcal{G} = (\mathcal{V}, \mathcal{E}, W)$ over the N candidates:

1. *Pairwise distance:* Compute $d_{ij} = d_{\text{PQC}}(\mathcal{P}_i, \mathcal{P}_j)$ for all pairs, or use VT-Aligned distance.

2. *KNN sparsification:* For each node i, retain edges to its $K = 3$ nearest neighbors.

3. *Edge weights:* $W_{ij} = \exp(-\beta \cdot d_{ij})$, with $\beta = 2.0$.

4. *Symmetrization:* If $j \in \mathcal{N}(i)$ but $i \notin \mathcal{N}(j)$, add the reverse edge with $W_{ji} = W_{ij}$.

Initial scores. Each document receives an initial "concentration":

$$C_0(i) = \exp(-2 \cdot d_{\text{PQC}}(\mathcal{P}_q, \mathcal{P}_i)) \tag{10}$$

Laplacian smoothing iteration. The score update follows the discrete heat equation on the graph:

$$C_{t+1}(i) = C_t(i) + \alpha \sum_{j \in \mathcal{N}(i)} W_{ij} \left(C_t(j) - C_t(i)\right) \tag{11}$$

where $\alpha = 0.15$ is the diffusion coefficient. $T = 5$ iterations are used. This can be written in matrix form as:

$$C_{t+1} = (I - \alpha L)\, C_t \tag{12}$$

where $L = D_{\text{deg}} - W$ is the graph Laplacian and D_{deg} is the degree matrix.

Stability condition. The eigenvalues of the update matrix $(I - \alpha L)$ lie in $[0, 1]$ iff $\alpha \leq 2/\lambda_{\max}(L)$, where $L = D - W$ is the graph Laplacian. For weighted graphs, $\lambda_{\max}(L) \leq 2d_{\max}^{(W)}$ where $d_{\max}^{(W)} = \max_i \sum_j W_{ij}$ is the maximum weighted degree (this is tight for bipartite graphs; otherwise loose). The sufficient condition $\alpha \leq 1/d_{\max}^{(W)}$ follows from $\lambda_{\max}(L) \leq 2d_{\max}^{(W)}$. Since edge weights $W_{ij} = \exp(-\beta d_{ij}) \in (0, 1]$, we have $d_{\max}^{(W)} \leq d_{\max}^{(\text{unweighted})}$, the unweighted degree. For $K = 3$ with bidirectional symmetrization, the unweighted degree is at most $2K = 6$, so $\alpha \leq 1/6 \approx 0.167$ suffices. Our choice $\alpha = 0.15$ satisfies this sufficient condition. Empirically the tighter bound $\alpha \leq 2/\lambda_{\max}(L)$ is rarely binding because $\lambda_{\max}(L) < 2d_{\max}^{(W)}$ on most KNN graphs we observed.

Effect. Laplacian smoothing propagates high relevance scores to semantically similar neighbors, effectively discovering relevant documents that may have received low initial scores due to vocabulary mismatch or embedding noise. Unlike cross-encoder reranking, this propagation is training-free and operates in milliseconds.

3.4. Multi-Granularity Pipeline

Our full retrieval pipeline operates in three stages with complementary granularities.

Stage 0: Centroid coarse filtering. Compute cosine similarity between the query centroid $\bar{v}_q$ and all document centroids $\{\bar{v}_i\}$ in the corpus. Retain the top-$K_1 = 100$ candidates. This stage runs in 23 ms on commodity CPU hardware.

Stage 1: Token PQ-Chamfer reranking. For the top-100 candidates, compute the full token-level PQ-Chamfer distance $d_{\mathrm{PQC}}(\mathcal{P}_q, \mathcal{P}_i)$ using all token hidden states. Re-rank and retain the top-$K_2 = 55$ candidates. This stage runs in 23 ms.

Stage 2: Graph Laplacian smoothing. Construct a KNN graph over the top-55 candidates and apply 5 iterations of Laplacian smoothing as described in Section 3.3. This stage runs in 26 ms.

Score fusion. For datasets where token-level data is available (NFCorpus, SciFact, ArguAna, SCIDOCS, FiQA), the two signal sources are combined:

$$\mathrm{score}_{\mathrm{final}}(i) = (1 - \lambda) \cdot \mathrm{score}_{\mathrm{token}}(i) + \lambda \cdot \mathrm{score}_{\mathrm{graph}}(i) \tag{13}$$

where $\lambda \in [0, 1]$ trades off pure token matching ($\lambda = 0$) against pure graph propagation ($\lambda = 1$). For Table 3 we report the per-dataset optimum $\lambda^\star$ on all five datasets with token-level data (NFCorpus $\lambda^\star = 0.4$, ArguAna $\lambda^\star = 0.0$, SciFact $\lambda^\star = 0.7$, SCIDOCS $\lambda^\star = 0.2$, FiQA $\lambda^\star = 0.2$). Both score vectors are min-max normalized before fusion. The dataset-dependence of $\lambda^\star$ and complete λ sweep are analyzed in detail in Section 4.7.

Two-stage acceleration is lossless. The centroid coarse filter reduces the candidate set from the full corpus to 100 documents before token-level scoring. This 13× speedup (305 ms full-scan to 23 ms two-stage) introduces *zero* accuracy loss: the two-stage pipeline achieves NDCG@10 = 0.3220 versus 0.3214 for the full token scan, with the slight improvement attributable to the centroid filter's implicit regularization effect.

Complexity analysis. Let N denote corpus size, m query tokens, n average document tokens, $M = 64$ subspaces, $K_1 = 100$ coarse candidates, and $K_2 = 55$ rerank candidates.

- *Stage 0 (centroid coarse filtering):* $O(N \cdot d)$ cosine comparisons, or $O(N \cdot m \cdot M)$ if using PQ-Chamfer at centroid level.

- *Stage 1 (token PQ-Chamfer reranking):* $O(K_1 \cdot m \cdot n \cdot M)$ per query—for each of K_1 candidates, compute $m \times n$ token-pair distances across M subspaces.
- *Stage 2 (graph Laplacian smoothing):* $O(K_2^2 \cdot n^2 \cdot M)$ for pairwise graph construction $+ O(T \cdot K_2 \cdot K)$ for T smoothing iterations on a K-sparse graph.

With typical NFCorpus parameters ($m \approx 6$, $n \approx 356$, $K_1 = 100$, $K_2 = 55$, $M = 64$), the bottleneck is Stage 1 at ∼137M floating-point operations per query, completed in 23 ms on commodity CPU.

4. Experiments

4.1. Experimental Setup

Datasets. We evaluate on six datasets from the BEIR benchmark [37], selected to cover diverse domains, corpus scales (2K–523K documents), document lengths (60–1,700 characters), and retrieval tasks. Table 2 summarizes the key statistics.

These six datasets test our pipeline under systematically varying conditions: corpus scale spans three orders of magnitude (2K to 523K), average document length ranges from 60 characters (Quora) to 1,700 characters (NFCorpus), and domains range from specialized (biomedical, financial) to general (community QA). This diversity ensures that observed improvements reflect genuine properties of the geometric operations rather than domain-specific artifacts.

Token-level data availability. Token-level embeddings were extracted for five of six datasets. Quora token-level extraction was not performed due to storage constraints: 522,931 documents × ∼170 tokens/doc × 4096 dimensions × 4 bytes ≈ 1.3 TB, exceeding available storage. Quora results use sentence-level graph smoothing only.

Embedding model. Qwen3-8B (Q4_K_M quantization) via llama.cpp with `-pooling none` for token-level hidden states and `-pooling mean` for centroid vectors. Embedding dimension $d = 4096$. Q4_K_M quantization was used for VRAM efficiency (8 GB usage vs. ∼16 GB for fp16); a full-precision (fp16) sanity check on the same model and data to bound the quantization-induced NDCG@10 deviation is deferred to a follow-up. Documented Q4_K_M perplexity overhead on standard benchmarks is ∼1–3% over fp16 (per llama.cpp's published benchmarks at `https://github.com/ggerganov/`

`llama.cpp/blob/master/examples/quantize/README.md`), which is small relative to the +6.0% (Quora, PDE-only) to +157.4% (FiQA, GBM-11) NDCG@10 gains reported in Table 3; we therefore expect the quantization to be a second-order effect on the retrieval-quality conclusions of this paper but acknowledge that a directly measured fp16-vs-Q4 contrast is not yet available.

PQ parameters. $M = 64$ subspaces, $d_s = 64$ dimensions per subspace, 256 centroids per subspace (for PQ index construction).

Graph parameters. KNN with $K = 3$, edge weight $W_{ij} = \exp(-2 \cdot d_{ij})$, Laplacian smoothing $\alpha = 0.15$, $T = 5$ iterations.

Cosine baseline conventions. Three distinct cosine similarities appear in this paper, all using the same Qwen3-8B model but with different pooling/candidate-pool conventions:

(S) *Sentence-level centroid cosine*: pooled mean over the whole document, query against full corpus. This is the canonical baseline reported in Table 3 (e.g., SciFact 0.4483, FiQA 0.1683).

(T) *Token-level centroid cosine*: pooled mean over per-token hidden states, query against full corpus. Used in Tables 11, 13, 12 as the first-stage for token-level fusion experiments (e.g., SciFact 0.4701, FiQA 0.1683 on the same query/corpus pool).

(T,K=55) *Token-level cosine restricted to the top-55 candidate pool*: same as (T) but evaluated on the 55-candidate pool produced by Stage 0 (centroid coarse) + Stage 1 (token PQ-Chamfer). Used in Table 11 as the $\lambda = 0$ column anchor (e.g., SciFact 0.456, NFCorpus 0.322).

The three numbers can differ because (T) uses token-mean (more granular than sentence-mean) and because (T,K=55) is a subset of the full corpus. **All NDCG@10 "gain" percentages in Table 3 use convention (S)** as denominator; **all NDCG@10 "gain" percentages in ablation tables use the matched first-stage cosine**. Each table caption identifies the cosine convention used.

Baselines. (1) Cosine similarity (single-vector retrieval), (2) BM25 (sparse retrieval), (3) an earlier-iteration convection-diffusion baseline on sentence-level point clouds (de-

noted in our internal logs as a prior PDE-style approach). We also contextualize our results against published numbers from trained retrieval models (ColBERTv2, BGE-large-en-v1.5) and LLM-based rerankers (RankGPT [36], RankLLM [28]). LLM-based rerankers such as RankGPT achieve state-of-the-art results on BEIR benchmarks but require expensive API calls ($0.01–0.10 per query); AFR-Rank [24] mitigates this cost through noise document filtering. Our pipeline—training-free geometric operators with a supervised per-query fusion-weight predictor—achieves competitive results at orders of magnitude lower cost ($47 total for all experiments).

Metric. NDCG@10, the standard metric for BEIR evaluation. Statistical significance is assessed via paired bootstrap with 10,000 iterations.

4.2. Main Results

Table 3 presents the main results across six BEIR datasets, using a single consistent decision rule for our pipeline output: the supervised **GBM-11** adaptive-fusion predictor (Section 4.8), which selects λ from per-query proxies in a 50/50 train/test split averaged over 5 seeds. This replaces the per-corpus oracle $\lambda^\star$ used in earlier versions of this work, which conflated test-set hyperparameter selection with model output. Component-level breakdowns (Cosine / $\text{Lap}_{K=55}$ / token_2stage / oracle upper bound) are presented separately as an ablation in Table 4.

Three patterns are observed under the *same-encoder Qwen3-8B-Q4 configuration* (cross-encoder transfer is discussed separately in Section 4.13 and bounded in Section 5.3). First, token-level PQ-Chamfer is the dominant component, providing large improvements on 4 of 5 datasets (+45.0% to +131.5%) with the sole exception of SciFact (−3.1%), where the strong cosine baseline (0.4483) limits available headroom. Second, graph Laplacian smoothing is the most consistent single improvement *within the same-encoder Qwen3 regime*, outperforming cosine on all six BEIR datasets evaluated (NFCorpus, ArguAna, SciFact, SCIDOCS, FiQA, and Quora; +5.8% to +44.2%); cross-encoder transfer is more nuanced and requires per-(model, corpus) tuning (Section 4.13). Third, score fusion helps selectively—it improves FiQA, SCIDOCS, and NFCorpus but hurts ArguAna, where token matching alone captures the dominant relevance signal. Pure Laplacian smoothing also outperforms the full convection-diffusion PDE on 5 of 6 datasets, which is the basis for retiring the advection formulation (Section 4.12).

Comparison with trained models. The pipeline outperforms BGE-large-en-v1.5 on FiQA and SCIDOCS, approaches ColBERTv2 on ArguAna (0.4418 vs 0.460) and NFCorpus (0.3270 vs 0.338), and exceeds BM25 on ArguAna and NFCorpus. Gaps remain where retrieval-specific training provides the most benefit (SciFact: 0.495 vs BGE 0.738; Quora: 0.675 vs BGE 0.889). Cross-model experiments (Section 4.13) confirm these gains generalize across embedding models. Implications are discussed in Section 6.

4.3. Statistical Significance and Identical-Split Baselines

Statistical significance is reported on two reference configurations.

(a) Per-corpus oracle $\lambda^\star$ reference. Paired bootstrap (10,000 iterations, seed 42) over per-query NDCG@10 compares *Oracle Best* (per-corpus $\lambda^\star$, Table 4) against *Cosine* on all five BEIR datasets with token-level data. The resulting two-sided p-values are robust under 51-seed replication. **Four of five corpora** (NFCorpus, ArguAna, SCIDOCS, FiQA) report $p < 0.001$ with $p_{\max}$ across 51 seeds $= 0.0001$. **SciFact reports** $p < 0.005$ with $p_{\max} = 0.0014$, the only corpus where the single-seed $p < 0.001$ claim is borderline. Cohen's d ranges from 0.197 (SciFact) to 0.717 (FiQA) and is stable across seeds (per-corpus d standard deviation < 0.001). Per-corpus ΔNDCG@10 with 95% CI and d: NFCorpus +0.0970 ($[+0.078, +0.117]$, $d = 0.552$); ArguAna +0.1370 ($[+0.121, +0.153]$, $d = 0.443$); SciFact +0.0462 ($[+0.020, +0.073]$, $d = 0.197$); SCIDOCS +0.1072 ($[+0.095, +0.119]$, $d = 0.562$); FiQA +0.2293 ($[+0.205, +0.254]$, $d = 0.717$). The 51-seed robustness check (50 seeds 0–49 plus seed 42) rules out single-seed bootstrap volatility as a confounder. Implementation: `benchmark/paired_bootstrap_5corpora.py` (single-seed) and `benchmark/paired_bootstrap_robustness_50seed.py` (51-seed verification); full outputs: `benchmark/data/results/paired_bootstrap_5corpora.json` and `paired_bootstrap_robustness_50seed_20260503.json`.

(b) GBM-11 deployment-time reference. Under the supervised GBM-11 fusion-weight predictor (5-seed train/test 50/50 split mean), per-corpus NDCG@10 lies within 4.0–6.1% of the per-corpus 5-seed-mean oracle upper bound on each of the 5 corpora (Table 4). Macro-mean GBM-11 (0.4072) is +11.5% relative to the fixed-λ baseline and −4.4% relative to the 5-seed-mean oracle upper bound (0.4261). Significance

on the GBM-11 reference is reported per-seed: the per-seed paired-bootstrap p-value of GBM-11 vs. Cosine is uniformly $< 10^{-3}$ across all 5 seeds on each of the 5 corpora. The 5-seed-mean point estimate is therefore a strictly less-noisy summary than any single-split number. Per-query rescue rates (Table 10) confirm the gain is not concentrated in a small minority. Quora is excluded from this test as it lacks token-level per-query data.

Re-evaluated baselines on identical splits. Four baselines are re-evaluated on identical splits in our own infrastructure (Table 6): ColBERTv2 (110M, BERT-based multi-vector with retrieval training); BGE-M3 (568M, Q8-quantized single-vector dense embedding); BGE-reranker-v2-m3 (568M, cross-encoder reranker on BGE-M3 cosine top-100 candidates); and a Qwen3-8B-Q4 LLM-listwise reranker (RankGPT-style [36]; see Section 4.9 and Table 13). Pairwise and setwise reranking with open-source LLMs are reported in Tables 14 and 15 (5 corpora full coverage). E5-Mistral-7B-Instruct reproduction was attempted but failed across three platforms and three dtypes (NFCorpus NDCG@10 $\sim$ 0.13–0.14 vs published $\sim$ 0.39); we document this in Section 4.10 as a failed-reproduction record (not a baseline result) and *exclude* E5-Mistral from the main baseline comparison. PLAID's centroid-pruning index requires CUDA-optimized C++ extensions that conflict with our ROCm 7.2 + `transformers` ≥ 5.7 stack (Section 4.11). Since PLAID's NDCG@10 differs from standard ColBERTv2 by sub-0.5% [34, Table 5], our ColBERTv2 numbers represent the same retrieval-quality data point.

4.4. Same-Infrastructure Ranking under Audited Rebuild + GBM-11 Deployment Rule

Across the five non-Quora datasets, the ranking comparison is reported under *two protocol lenses* to forestall a protocol-mismatch concern.

Lens 2 (protocol-matched, single-split, primary). Using the single-split single-seed Ours numbers reported in the rightmost column of Table 6 (0.3270/0.4946/0.4418/0.2147/0.3977 on NFCorpus/SciFact/ArguAna/SCIDOCS/FiQA), evaluated under the identical single-split full-evaluation protocol as the four audited baselines, our pipeline ranks *first* on one corpus (SCIDOCS, 0.2147 vs. next-best BGE-reranker 0.1823, +17.8%); *second* on three (NFCorpus 0.3270 vs. BGE-reranker 0.3275, −0.15%; ArguAna 0.4418 vs. BGE-reranker 0.4993, −11.5%; FiQA 0.3977 vs. BGE-reranker 0.4227, −5.9%); and *fourth* on SciFact

(0.4946, behind ColBERTv2 0.6141, BGE-M3 0.6406, BGE-reranker 0.7228). This is the strict protocol-matched comparison (Ours and baselines all on the same single-split single-seed full evaluation), and we adopt it as the primary headline reading.

Lens 1 (deployment-time, 5-seed 50/50, robustness cross-check). Our pipeline (GBM-11 fusion, the Table 3 *Ours* column, 5-seed train/test 50/50 split mean) ranks *first* on three corpora: NFCorpus (0.3297 vs. next-best BGE-reranker-v2-m3 audited 0.3275, +0.7%), SCIDOCS (0.2451 vs. 0.1823, +34.4%), and FiQA (0.4331 vs. 0.4227, +2.5%). It ranks second on ArguAna (0.4862 vs. 0.4993, −2.6%). It ranks third on SciFact (0.5418 vs. ColBERTv2 in-house 0.6141 and BGE-reranker-v2-m3 audited 0.7228, −11.8% and −25.0% respectively). *Caveat*: this lens compares 5-seed-mean Ours against single-split single-seed audited baselines (Table 6); it is the deployment-rule reading and is *not* strictly protocol-matched, hence reported here as a robustness cross-check rather than as the primary number.

(c) Per-seed dispersion under Lens 1 (direct evidence for the less-noisy-point-estimate claim). The Lens 1 vs. Lens 2 first-rank divergence (3 vs. 1) is justified above on the methodological ground that the 5-seed-mean point estimate is strictly less noisy than a single-split single-seed estimate. Direct dispersion evidence is summarised in Table 5: per-seed standard deviation of NDCG@10 across the 5 train/test 50/50 split seeds, and the per-seed empirical tally (5-seed Ours-GBM-11 vs. the next-best single-split single-seed audited baseline). The tally is consistent with the Lens 1 first-rank reading on SCIDOCS (5/5 dominance), borderline on NFCorpus (3/5) and FiQA (4/5), and dominated by the audited baseline on ArguAna and SciFact (0/5); on ArguAna and SciFact, no per-seed first-rank is claimed under either lens.

Lens summary. The two lenses agree on (i) SCIDOCS as the strongest first-rank result, (ii) SciFact as the primary weak-baseline-boundary case, and (iii) the boundary-positioning conclusion that retrieval-trained cross-encoders dominate on short scientific abstracts. They disagree on first-rank count (Lens 2: 1 on SCIDOCS; Lens 1: 3 on NFCorpus/SCIDOCS/FiQA): the 5-seed-mean point estimate recovers signal margin lost in single-split single-seed evaluation. Lens 2 is the primary headline (protocol-matched against single-split single-seed audited baselines); Lens 1 is the deployment-rule cross-check (the lens under which GBM-11 is itself defined; Table 4 footnote and Section 4.8).

Per-seed first-rank consistency under Lens 1 is reported in subsection (c). On SciFact, neither lens reaches the absolute top tier (consistent with the boundary-positioning thesis, Section 6.1). The present submission applies the GBM-11 deployment rule (Table 3) and audited-rebuild baselines (Table 6) after resolving two methodological issues: oracle $\lambda^\star$ replaced by held-out 50/50 split; BGE-reranker-v2-m3 first-stage rebuilt to remove 19.93%/18.62% zero-encoded NFCorpus/SciFact documents.

4.5. Ablation Study (NFCorpus)

Table 7 presents a detailed ablation on NFCorpus, the dataset for which we have complete token-level data.

Statistical significance. Token two-stage vs cosine baseline: paired bootstrap with 10,000 iterations, $p < 0.001$. Fusion vs token two-stage: $p = 0.3346$[2] (not significant), indicating that the graph smoothing component provides a modest but non-significant additive benefit on this dataset.

Progression analysis. The ablation reveals a clear hierarchy of contributions:

- *Sentence-level point cloud* (early Chamfer iteration, +24.3%): Moving from single-vector to multi-sentence representation provides substantial gains.
- *PQ subspace decomposition* (sentence-level PQ-Chamfer iteration, +24.3% → +27.4% via the VT-Aligned token-level iteration): Breaking the high-dimensional space into subspaces further improves discrimination.
- *Token-level point cloud* (the present token-level iteration, +46.7%): The jump from sentence-level to token-level representation is the single largest improvement, nearly doubling the gain from +24.3% to +46.7%.
- *Graph smoothing* (+32.1% standalone, +49.0% in fusion): Graph propagation provides an orthogonal signal that complements token matching.

[2]Computed at $\lambda^\star = 0.4$ (per-dataset NFCorpus optimum, 0.6× token + 0.4× graph) over 10,000 paired bootstrap iterations; the marginal gain over token-only (NDCG@10 0.3270 vs. 0.3220) does not reach 5% significance.

4.5.1. Multi-Dataset Pipeline Decomposition

Table 8 extends the ablation across all five datasets with token-level data, decomposing the pipeline into its constituent stages.

The cross-dataset decomposition confirms the patterns from Table 3. Token-level matching dominates on 4 of 5 datasets; SciFact is the exception because its strong cosine baseline (0.4483) already captures sufficient discriminative signal. Gains are inversely correlated with baseline strength: weaker baselines see larger improvements. Fusion is regime-specific—it helps FiQA, SCIDOCS, and NFCorpus but hurts ArguAna (-4.6%), and the optimal configuration is dataset-dependent. FiQA's single-chunk documents, which consist of a single embedding vector each, still benefit from PQ subspace decomposition because PQ creates an implicit point cloud of 64 independent projections.

4.6. Summary of Findings

Three key findings emerge. First, token-level PQ-Chamfer (oracle fusion column in Table 4) yields $+45\%$ to $+136\%$ gains on 4/5 datasets under the same-encoder configuration, with gains inversely correlated with cosine baseline strength (not document length). FiQA's single-vector documents achieve the largest improvement via PQ's implicit "virtual token" decomposition. Second, the two scoring signals exhibit dataset-dependent complementarity (Kendall τ ranging from 0.31 to 0.55, see Section 4.7 and Table 10); fusion helps NFCorpus and SciFact but not ArguAna, where token PQ-Chamfer alone saturates the signal at NDCG@10 $= 0.442$. Third, under the deployment GBM-11 fusion rule (Table 3), against the same-infrastructure audited-rebuild BGE-large reference (the fair within-pipeline comparison standard) the pipeline surpasses BGE-large on FiQA (0.4331 vs. audited-rebuild BGE-large 0.385, $+12.5\%$); the published BGE-large-en-v1.5 reference number on FiQA is 0.367, against which the same Ours-GBM-11 number gives $+18.0\%$ (the published-vs-audited gap reflects inference-framework and truncation differences documented in the *Implementation note* of Section 4.13, not pipeline gain). On SCIDOCS, Ours-GBM-11 attains 0.2451 vs. published BGE-large-en-v1.5 0.162 ($+51.3\%$); the same-infrastructure audited-rebuild BGE-large SCIDOCS reference is not separately re-encoded in this submission (the BGE-large vector budget was prioritized for NFCorpus/FiQA), so this reduction-to-published is reported with the framework-difference caveat. Together these demonstrate that training-free geometric operations (composed

with the supervised GBM-11 per-query fusion-weight rule of Section 4.8) can recover sub-document structure that contrastive training does not fully exploit, with the audited-rebuild reference taken as the primary fair baseline and the published reference reported as a secondary cross-check.

4.7. Fusion Mechanism Analysis

Concerns previously raised about the fusion of PQ-Chamfer and graph Laplacian smoothing fall under three questions: (i) the optimal number of smoothing iterations, (ii) the apparent paradox between component complementarity and fusion outcome, and (iii) the choice of fusion weight λ. We address each.

(i) Iteration count T.. The graph Laplacian iteration $C^{(t+1)} = (I - \alpha L)C^{(t)}$ (Eq. 11 in matrix form, equivalent to $C^{(t+1)}(i) = C^{(t)}(i) + \alpha \sum_j W_{ij}(C^{(t)}(j) - C^{(t)}(i))$) has a single dominant timescale controlled by the spectral gap $\lambda_2(L)$ of the graph Laplacian. Table 9 reports a sweep on NFCorpus, ArguAna, and SciFact (extended on NFCorpus to $T \in \{25, 50, 100, 200\}$). The peak is consistently at $T = 3$, only 1.1% above the manuscript default $T = 5$ across all three datasets. Past $T = 20$, NFCorpus exhibits sharp monotone degradation: at $T = 200$, NDCG@10 falls to 0.171, 22% below the cosine baseline (0.2195, consistent with Table 3 full-eval cosine; the value 0.230 in fusion_ablation_results.json reflects the $K = 55$ candidate pool truncation). This is direct empirical evidence that diffusion past the spectral-gap timescale homogenizes the score field and destroys discriminative signal. The default $T = 5$ in our pipeline is near-optimal but not at the precise peak; in production, $T = 3$ is preferable.

(ii) Complementarity vs fusion paradox.. Table 10 reports four complementarity measures across the three datasets. Kendall τ varies from 0.31 (NFCorpus) to 0.55 (ArguAna), confirming that rank disagreement between PQ-Chamfer and Laplacian rankings is non-trivial across datasets. Yet fusion outcomes do not track τ: ArguAna has the *highest* τ but the *worst* fusion gain (negative for any $\lambda > 0$). This is the diagnostic counter-example. Kendall τ is a rank-correlation statistic that does not predict additive complementarity in score space. Two signals can disagree on rank order yet produce calibrated scores whose convex combination is dominated by either component alone—the

relevant diagnostic is per-query score-distribution overlap, not rank correlation. ArguAna's token PQ-Chamfer alone reaches NDCG@10 = 0.442 vs cosine baseline 0.305 (a +45% gain), saturating the available signal; mixing in any Laplacian contribution dilutes calibrated token scores. The precise statement is therefore: rank disagreement is necessary but not sufficient for fusion gain.

(iii) Fusion weight λ.. The fusion score $s_{\text{fuse}} = (1-\lambda)s_{\text{PQ}} + \lambda s_{\text{Lap}}$ with the fixed-λ default $\lambda = 0.3$ is swept over $\lambda \in \{0, 0.1, \ldots, 1.0\}$ on all five BEIR datasets with token-level data in Table 11. The optimal $\lambda^\star$ varies sharply across datasets: NFCorpus $\lambda^\star = 0.4$ (NDCG@10 0.327), ArguAna $\lambda^\star = 0.0$ (token alone strictly dominates—any positive λ hurts), SciFact $\lambda^\star = 0.7$ (NDCG@10 0.495), SCIDOCS $\lambda^\star = 0.2$ (NDCG@10 0.218), FiQA $\lambda^\star = 0.2$ (NDCG@10 0.398). Consequently no single global λ is justified; the fixed-λ default $\lambda = 0.3$ is suboptimal on four of five datasets (only NFCorpus's optimum at 0.4 is close).

4.8. Adaptive Fusion: Supervised Per-Query λ Predictor

The dataset-dependent $\lambda^\star$ in Table 11 motivates a query-adaptive rule that replaces the manually-set fusion weight (addressing earlier feedback on score-fusion adaptivity). We instantiate two predictors trained to map per-query proxy features to a per-query $\hat{\lambda}$.

Ridge-4 baseline (Ridge regression, 4 features). Ridge regression over four features: Kendall τ, Spearman ρ, cosine-baseline NDCG@10, and query length.

GBM-11 (gradient-boosted regression, 11 features). Gradient-boosted regression trees (200 estimators, depth 3, learning rate 0.05) over eleven features: the four Ridge-4 features plus (i) `token_2stage` NDCG@10, (ii) `lap_T5` NDCG@10, (iii) token–cosine improvement margin, (iv) lap–cosine improvement margin, (v) `lap_T1`−`lap_T20` spread, (vi) $|\texttt{token} - \texttt{lap}|$ disagreement, and (vii) lap T-sweep range max − min.

Both predictors are trained on the per-query oracle $\lambda^\star$ in a 50/50 train/test split with the test half held out, averaged over 5 random seeds.

Macro results. Across all five BEIR corpora with token-level data (NFCorpus, ArguAna, SciFact, SCIDOCS, FiQA; total 3,669 test queries), the macro NDCG@10 progresses as follows. Fixed $\lambda = 0.3$ achieves 0.3653. Per-corpus $\lambda^\star$ (chosen on the train half, applied to the test half) achieves 0.3711. Ridge-4 achieves 0.3769 (+3.18%

relative-to-fixed). **GBM-11 achieves 0.4072** (+**11.5**% relative-to-fixed). The 5-seed-mean oracle upper bound is 0.4261. The GBM-11-to-oracle gap is 0.0189 absolute (5.17% relative-to-fixed), a **2.6**× closing of the Ridge-4 gap of 13.5%. SciFact's Ridge-4 under-performance (0.4724 vs. fixed 0.4771) is recovered by GBM-11 (0.5418), confirming that the richer feature set plus non-linear model dissolves the overfitting concern visible in Ridge-4.

Implementation and reproducibility. `benchmark/adaptive_fusion_lambda_v2.py` (training script); `adaptive_lambda_v2_gbm_results.json` (per-corpus 5-seed JSON output). The same script supports a single-seed re-run for ablation purposes.

Online deployment-time extension (proof-of-concept). The GBM-11 predictor is offline-trained on per-query oracle $\lambda^\star$ targets, which is a supervised-learning convenience and not a deployment-time policy: at deployment we do not have access to oracle $\lambda^\star$ per query. We therefore additionally implement a contextual-bandit online policy that does not require any oracle label at query time. We use disjoint LinUCB [22] with the same eleven per-query features as context, eleven discrete arms ($\lambda \in \{0.0, 0.1, \ldots, 1.0\}$), and per-query NDCG@10 as the bandit reward. The policy is updated sequentially: query t observes $(\mathbf{x}_t, a_t, r_t)$, applies a Sherman–Morrison rank-one update to the per-arm $\mathbf{A}_a^{-1}$ posterior, and selects a_{t+1} via the LinUCB acquisition $\arg\max_a[\boldsymbol{\theta}_a^\top \mathbf{x} + \alpha\sqrt{\mathbf{x}^\top \mathbf{A}_a^{-1}\mathbf{x}}]$. On NFCorpus (323 queries, 5 shuffle-and-tie-break seeds, exploration coefficient $\alpha = 0.5$), the LinUCB online policy achieves a macro NDCG@10 of 0.3294 ± 0.0023, which matches the GBM-11 offline-supervised reference of 0.3297 (-0.09% relative gap) and surpasses fixed $\lambda = 0.3$ (0.3264) by +0.90%. The cumulative regret over 323 queries is 10.85 ± 0.74 NDCG units (average per-query regret 0.034), and the running 50-query NDCG@10 mean stabilises after roughly $N \approx 100$–150 queries. The remaining gap to the per-query oracle ($0.3630 - 0.3294 = 0.0336$) is comparable to the gap from fixed $\lambda = 0.3$ to oracle (0.0365), so on this corpus the online policy primarily recovers the GBM-11 offline performance *without requiring oracle labels at deployment time*, rather than further closing the oracle gap. We additionally retain the legacy supervised replay variant (`benchmark/adaptive_lambda_online.py`, incremental SGD with $\eta = 0.005$ and CUSUM drift threshold $h = 2.0$): replaying the 3,669 test queries through the SGD update yields macro NDCG@10 between GBM-11 (0.4072)

and the 5-seed oracle (0.4261), but this replay variant assumes oracle $\lambda^\star$ as the supervised target and is therefore not a deployment-time policy. The LinUCB result above is, by contrast, a real online policy: it observes only the realised reward $r_t = \text{NDCG@10}$ of its own action, never the oracle $\lambda^\star$. This narrows the residual concern raised in earlier feedback that the score-fusion strategy was not consistent and lacked a stable adaptive mechanism, to a residual concern about the asymptotic regret bound, multi-corpus generalisation, and a full-pipeline online integration with click-stream feedback, which we leave to future work. Implementation: `benchmark/online_linucb_toy.py`; results: `benchmark/data/results/online_linucb_nfcorpus_results.json`. this is a proof-of-concept on a single corpus (NFCorpus), with simplifying assumptions (linear reward model per arm, discrete eleven-arm grid); a multi-corpus deployment study and a regret-bound analysis with finite-time guarantees are deferred.

Remaining gap. The $\sim 4.4\%$ oracle gap suggests two avenues for further reduction: (i) feature interactions (current GBM is depth-3, suggesting richer cross-feature splits); (ii) cross-encoder calibration signals (e.g., BGE-reranker top-1 logit) as an additional feature. Both extensions are sketched but not implemented in the present submission. **Hartree-style self-consistent λ variant (training-free, two free parameters; negative result on NFCorpus).** The GBM-11 predictor above is a *post-hoc supervised regressor*: it requires per-query oracle $\lambda^\star$ targets computed from `qrels` on the training half, eleven hand-crafted features, and a 50/50 train/test split. As a forward, training-free alternative we additionally implement a Hartree-style self-consistent fixed-point iteration (a standard mean-field-theory closure used in many-body physics; 14): the per-query effective fusion weight $\lambda^{\text{eff}}(q)$ is defined as the fixed point of

$$\lambda^{\text{eff}}(q) \;=\; \lambda_0 \;+\; \lambda_\Sigma \cdot \left\| R(q;\, \lambda^{\text{eff}}) \right\|^2, \qquad R(q;\lambda) \;=\; (1-\lambda)\,\text{tok}(q) \;+\; \lambda\,\text{lap}(q), \tag{14}$$

where $\text{tok}(q)$ and $\text{lap}(q)$ are the per-query token-stage and lap-T5 NDCG@10 signals, and $(\lambda_0, \lambda_\Sigma)$ are the only two free parameters (versus eleven for GBM-11, a $5\times$ reduction). The fixed point is found by damped Picard iteration ($\alpha = 0.5$, tolerance 10^{-2}, typically converging in 4–10 iterations across 5 seeds; full trace in the result JSON). Crucially, no `qrels` information enters the iteration: the fixed point is determined entirely by the cosine + token + lap retrieval signals already produced by the geometric pipeline, making this variant a candidate *deployment-time* fusion rule. **NFCorpus re-**

sult. Under the same 5-seed 50/50 protocol as GBM-11, with $(\lambda_0, \lambda_\Sigma)$ tuned on the train half (NDCG@10 grid search; no $\lambda^\star$ supervision) and applied to the held-out test half, the Hartree variant reaches macro NDCG@10 0.3106 ± 0.0148, which matches fixed $\lambda = 0.3$ (0.3116 ± 0.0138) within one standard deviation but underperforms GBM-11 (0.3297 ± 0.0156) by 0.0191 absolute (a 5.8% relative shortfall). The oracle upper bound on NFCorpus is 0.3469 ± 0.0164. **Honest interpretation.** The negative result reads cleanly: $\mathrm{tok}(q)$ and $\mathrm{lap}(q)$ are themselves *retrieval-quality* signals on the realised candidate list, but they do not fully determine the optimal mixing weight $\lambda^\star(q)$, which depends on the relative shape of the two ranked lists rather than on their per-list quality alone. Eqn. (14) therefore collapses much of the per-query structure that GBM-11 exploits via its eleven features (Kendall τ, Spearman ρ, lap T-sweep variance, signal disagreement). The supervised oracle $\lambda^\star$ supervision in GBM-11 captures structure beyond what self-consistent fixed-point iteration on retrieval-norm proxies recovers; in particular the lap T-sweep variance and the token–lap disagreement features carry signal that cannot be reconstructed from $R(q; \lambda)$ alone. We report this single-corpus negative result as evidence that the $11 \to 2$ free-parameter reduction is achievable in form (Eqn. (14) is a closed self-consistent equation), but *not at the same retrieval-quality level* as the supervised regressor; closing the gap would require either (i) a richer per-query closure than the second-moment Hartree term, or (ii) hybrid supervision that injects partial $\lambda^\star$ structure while retaining the deployment-time fixed-point form. Both are deferred. **Implementation:** `benchmark/hartree_lambda_toy.py`; result JSON: `benchmark/data/results/hartree_lambda_nfcorpus.json`; full 5-seed iteration trace with per-seed $(\lambda_0, \lambda_\Sigma)$ tuning and per-iter $\max_q |\Delta\lambda^{\mathrm{eff}}|$ included.

4.9. LLM-listwise Reranker Baseline

To address the earlier-feedback request for an explicit LLM-listwise reranker comparison (RankGPT [36], RankLLM [28]), we implement a RankGPT-style listwise reranker over the cosine top-20 candidates per query, run via a local Qwen3-8B-Q4_K_M model served by llama.cpp on the same RX 9070 XT (no API cost, $\sim$1.5 s/query latency). Qwen3's thinking mode is disabled via the `/no_think` prompt directive so the listwise prompt budget is not consumed by chain-of-thought tokens; outputs are constrained to a comma-separated permutation of bracketed candidate indices, parsed deterministically.

The first-stage cosine top-100 is the same Qwen3-8B token-level-centroid pool that defines our *Cosine* baseline (NFCorpus 0.2195), making this comparison a same-family rerank: the LLM and the first-stage embedding share the same parameters. Table 13 reports the result on the three datasets where token-level data is fully available (NFCorpus, SciFact, ArguAna).

† BGE-reranker uses a different (BGE-M3-based) first-stage; the negative "vs. BGE-rerank" entries reflect (i) Qwen3-8B's untrained token-centroid first-stage being weaker than BGE-M3 (e.g., 0.2195 vs. 0.3113 cosine NDCG@10 on NFCorpus, audited rebuild), and (ii) Qwen3-8B-Q4's listwise prompting being less calibrated than BGE-reranker-v2-m3's retrieval-trained cross-encoder.

Cross-first-stage disentanglement (5-corpus, audited). To separate effects (i) and (ii), we ran the same Qwen3-8B-Q4 listwise reranker on **BGE-M3 cosine top-100** (the same first-stage as BGE-reranker-v2-m3 in Table 6) across all 5 BEIR corpora using the audited-rebuild BGE-M3 vectors (see Section 4.13 *Vector-provenance note* + Table 6 *Audited rebuild note*):

- **NFCorpus**: BGE-M3 cosine 0.3113 → Qwen3-listwise **0.3330** (+6.97%, $n = 323$); BGE-reranker-v2-m3 audited 0.3275. *Indistinguishable* (Delta = +0.0055).
- **SciFact**: BGE-M3 cosine 0.6406 → Qwen3-listwise **0.6552** (+2.29%, $n = 300$); BGE-reranker-v2-m3 audited **0.7228**. BGE-reranker *stronger* (Delta = −0.0676, −10.3%).
- **ArguAna**: BGE-M3 cosine 0.3298 → Qwen3-listwise 0.2654 (−19.53%, $n = 1406$); BGE-reranker-v2-m3 audited **0.4993**. BGE-reranker *stronger* (Delta = −0.2339, −88%).
- **SCIDOCS**: BGE-M3 cosine 0.1676 → Qwen3-listwise **0.1896** (+13.16%, $n = 1000$); BGE-reranker-v2-m3 audited 0.1823. *Indistinguishable* (Delta = +0.0073).
- **FiQA**: BGE-M3 cosine 0.4038 → Qwen3-listwise 0.3803 (−5.81%, $n = 648$); BGE-reranker-v2-m3 audited **0.4227**. BGE-reranker *stronger* (Delta = −0.0424, −11.1%).

Single-corpus NFCorpus measurement (0.2591 → 0.3015 vs. BGE-rerank 0.3013, indistinguishable) does not generalize: the 5-corpus audited replication is the reference reading. The 5-corpus pattern is **task-dependent**: Qwen3-8B-Q4 listwise reranking is indistinguishable from BGE-reranker-v2-m3 on *NFCorpus and SCIDOCS*, but BGE-reranker-v2-m3 is stronger on *SciFact, ArguAna, and FiQA*. Generic LLM listwise reranking thus retains task-dependent strengths but does not replace retrieval-trained cross-encoders on every corpus. Implementation: `benchmark/llm_rerank_bench_bge.py` (BGE-M3 first-stage v2); results: `llm_rerank_results_bge_m3_*.json` (Qwen3-listwise) and `outputs/bge_reranker_*_eval_v2.json` (BGE-reranker-v2-m3 audited rebuild).

The finding (Qwen-centroid first-stage, Table 13) is that generic LLM listwise reranking is regime-specific to certain first-stages: it can help (NFCorpus, +24.3%), be neutral (SciFact), or sharply hurt (ArguAna, −17.1%). Across these three datasets, our pipeline (Table 6: NFCorpus 0.327, SciFact 0.495, ArguAna 0.442) outperforms LLM-listwise on each, with the largest absolute margin on ArguAna (+0.190 NDCG@10).

Pairwise reranker baseline (5-corpus, BGE-M3 first-stage, audited).. To address the earlier-feedback request for *pairwise* reranking using open-source LLMs (in addition to the listwise variant above and the setwise variant below), we use a Bradley–Terry-style pairwise reranker over BGE-M3 audited-rebuild top-20 candidates: for each query, B pairwise comparisons are sampled (Bradley–Terry pair-budget per query) and adjudicated by the same Qwen3-8B-Q4 model via a 2-document pairwise prompt; pairwise win counts plus first-stage score break ties to produce the final ranking. To make this baseline directly comparable with the setwise reranker (Table 15), we use pair-budget $B = 50$ per query for SciFact, ArguAna, SCIDOCS, and FiQA; the NFCorpus pair-budget was $B = 200$ from an earlier run. Implementation: `benchmark/llm_rerank_pairwise_setwise.py -mode pairwise -pair-budget {50,200}`; per-query JSONL: `llm_rerank_pairwise_setwise/pairwise_*.json`.

The pairwise reranker is intermediate in degradation between listwise (small mixed) and setwise (−29.5% macro): pairwise macro −12.2% shows that the more granular per-pair adjudication does extract some signal beyond a generic prompt-only setwise selection, but still degrades the BGE-M3 cosine baseline on 4 of 5 corpora. The two positive

cells (NFCorpus +15.4% at $B = 200$, ArguAna +1.2% at $B = 50$) are corpora where BGE-M3 cosine is relatively weak (NFCorpus 0.2895, ArguAna 0.3298) and pairwise discrimination has more headroom. The cross-baseline finding is consistent across all three open-source LLM rerank modes (listwise / pairwise / setwise): *generic LLM relevance prompting on top of an already-trained dense retriever does not reliably improve BEIR NDCG@10 in this regime*, while our geometric post-processing pipeline does (+11.2% macro Ours-GBM-11 vs. BGE-M3 cosine).

Setwise reranker baseline (5-corpus, BGE-M3 first-stage, audited).. To address the earlier-feedback request for *setwise* reranking based on open-source LLMs (in addition to the listwise variant above), we use a sliding-window setwise reranker over BGE-M3 audited-rebuild top-20 candidates: for each consecutive set of 4 candidates (50% overlap sliding window), the same Qwen3-8B-Q4 model picks the top-2 most relevant via a setwise prompt; promotion counts plus first-stage score break ties to produce the final ranking. The setwise prompt explicitly disables Qwen3's chain-of-thought via empty `<think></think>` tags, constraining output to a comma-separated 2-index pattern parseable in $\leq$ 16 tokens. Implementation: `benchmark/llm_rerank_pairwise_setwise.py -mode setwise -set-size 4`; per-query JSONL: `llm_rerank_pairwise_setwise/setwise_*.json`.

The *setwise* result is even stronger than the listwise result in degrading BGE-M3 cosine: -29.5% macro for setwise vs. a small mixed effect for listwise on the same first-stage. The 4-document setwise window appears to give Qwen3-8B-Q4 insufficient pairwise discrimination signal at the granularity that BGE-M3 cosine already captures, while introducing parsing-failure noise (parse failure rate 0.0–0.1% across corpora) and prompt-length-related calibration loss. The cross-baseline finding is that open-source LLM listwise/setwise reranking on top of an already-trained dense retriever (BGE-M3 cosine) does *not* reliably improve BEIR NDCG@10 in this regime, while our geometric post-processing pipeline does (+11.2% macro Ours-GBM-11 vs. BGE-M3 cosine). This is direct empirical evidence that geometric structural priors capture retrieval signal complementary to (and at this niche, more reliably effective than) prompt-based LLM relevance judgments. Pairwise reranker results on the same first-stage are reported above in Table 14; the three rerank modes (listwise / pairwise / setwise) are now all 5-corpus

complete.

4.10. E5-Mistral on AMD ROCm: Failed Baseline Reproduction

We were unable to reproduce E5-Mistral-7B-Instruct's published BEIR numbers within this submission's hardware/software stack. We document the reproduction attempt here but *exclude* E5-Mistral from the main baseline comparison in Table 6; this is a failed-reproduction record, not a baseline result. We attempted to evaluate E5-Mistral-7B-Instruct [38] as the LLM-based embedding baseline requested by RQ2. On the AMD RX 9070 XT (RDNA4, gfx1201) under ROCm with PyTorch 2.10, the model loads successfully and produces query/document embeddings, but the resulting NDCG@10 is anomalously low on every dataset we re-evaluated (NFCorpus 0.140, SciFact 0.305, ArguAna 0.083, SCIDOCS 0.044, FiQA 0.121, vs. published numbers ∼0.39, ∼0.71, ∼0.55, ∼0.16, ∼0.56 respectively).

Same-hardware native dtype control (audited rebuild). On RX 9070 XT under fixed encode parameters (ROCm 7.2, PyTorch 2.10.0+rocm7.2.0, `batch_q=1`, `batch_d=2`, `d_max_len=512`, `attn_implementation=sdpa`): native bf16 inference NDCG@10 = 0.1396 (n = 323); native fp16 inference NDCG@10 = 0.1392 (n = 323); difference Δ = 0.0005, within seed noise. Two dtype settings produce identical anomalously-low NDCG@10 on the same hardware, which *rules out* an isolated bf16 numerical issue. Configuration: `embeddings/e5_mistral_{bf16,fp16}_native_control/nfcorpus/meta.json`; per-platform JSON: `outputs/e5_mistral_nfcorpus_{bf16,fp16}_native_control_eval.json`.

Provenance audit note. An earlier draft of this paper cited the same-hardware dtype control with NDCG@10 ≈ 0.1401 (bf16) and ≈ 0.1392 (fp16) from runs on 2026-04-30 to 2026-05-01. A post-hoc internal audit (2026-05-01 evening) found two reproducibility issues with those original numbers: (i) the encode script's `meta.json dtype` field was a hardcoded string `"fp16 (computed in bf16, stored fp16)"` that did not reflect the runtime `-dtype` command-line argument; and (ii) the shell history of the original `-dtype` value was not retained. Both meta-level and provenance-level traces were therefore insufficient to verify the original dtype attribution. To close this gap we patched the encode script (`scripts/encode_e5_mistral_nfcorpus.py`: the `meta.json dtype` field now uses `args.dtype` dynamically) and rebuilt the control under explicit dtype tracking.

The rebuilt numbers (0.1396, 0.1392 above) confirm the earlier qualitative conclusion (two dtypes produce identical anomalously-low NDCG@10) and we use the audited rebuild numbers as the data points of record in this work.

Cross-platform CUDA control. To rule out the remaining "ROCm-platform-wide attention numerical bug" hypothesis, we additionally ran E5-Mistral on a NVIDIA RTX 5060 Laptop GPU (CUDA 13.1, torch 2.11+cu130) using bitsandbytes 4-bit nf4 quantization with fp16 compute (the only configuration that fits in 8 GB VRAM): NFCorpus NDCG@10 = **0.1289** ($n = 323$ test queries, encode time 18.8 min for 3,633 docs). Three platforms × three dtype configurations summary:

- RX 9070 XT, ROCm 7.2, native bf16+sdpa: 0.1396 (audited rebuild, $n = 323$)
- RX 9070 XT, ROCm 7.2, native fp16+sdpa: 0.1392 (audited rebuild, $n = 323$)
- RTX 5060, CUDA 13.1, nf4 4-bit + fp16 compute: 0.1289 ($n = 323$)

All three are within $[0.129, 0.140]$ — a 0.011 absolute spread well-matched within expected quantization noise (∼1–3% relative for nf4 vs fp16 per bitsandbytes documentation), and all anomalously low vs published ∼0.39.

Attribution analysis. The cross-platform control *rules out* (a) ROCm-specific platform bug (CUDA reproduces). The remaining hypothesis is (b) defect in our inference pipeline (tokenization, last-token pooling, normalization) or evaluation protocol (cosine on full corpus + top-10) that consistently underperforms the published E5-Mistral evaluation across hardware. The most likely candidates are: (i) last-token pooling implementation difference (left- vs right-padded inputs), (ii) the Instruct prompt template (we use the official template but field experience shows even minor whitespace / "Query:" formatting variation can shift NDCG by ∼0.05 on E5-Mistral), or (iii) our cosine-on-full-corpus evaluation not matching the published evaluation harness's first-stage retrieval scope. Localizing this is left to a follow-up. BGE-M3 (568M, Q8-quantized via llama.cpp) is used as the LLM-based embedding baseline of record for RQ2 in Table 6; the present section (Section 4.10) reports the failed-reproduction numbers separately. Implementation: `scripts/encode_e5_mistral_nfcorpus.py` (ROCm, audited rebuild script with dynamic `args.dtype` meta tracking), `benchmark/e5_cuda_control.py` (CUDA); per-platform JSON: `embeddings/e5_mistral_{bf16,fp16}_native_control/nfcorpus/`

and `e5_cuda_control/nfcorpus/e5_cuda_4bit_nfcorpus_eval.json`.

4.11. Reproducibility Notes and Deferred Comparisons

We list here three comparisons that are referenced in Tables 6 and 13 but not run on our hardware, with the explicit reason for deferral.

PLAID latency benchmarking.. PLAID [33] is the centroid-pruning efficiency layer over ColBERTv2 and is the standard high-throughput retrieval implementation for late-interaction models. Its NDCG@10 differs from standard ColBERTv2 by sub-0.5% on MS MARCO and several BEIR datasets in the original paper (Santhanam et al. 2022, Table 5); we accordingly report ColBERTv2 numbers (Table 6) as representative of this model family's retrieval *quality*.

Latency comparison (measured + published). We measured full-MaxSim ColBERTv2 latency (no PLAID centroid pruning) on our hardware (RX 9070 XT, ROCm 7.2): NFCorpus 288 ms/query, SciFact 1,281 ms/query, ArguAna 3,065 ms/query, SCIDOCS 4,251 ms/query, FiQA 9,733 ms/query (`outputs/colbertv2_*_eval.json`). PLAID's published latency on MS MARCO is ∼58 ms/query end-to-end (Santhanam et al. 2022, Table 6); on BEIR, PLAID is reported in subsequent benchmarks at ∼50–200 ms/query depending on corpus size. Our pipeline—training-free geometric operators with a supervised per-query fusion-weight predictor—incurs CPU-only latency of ∼50 ms/query at inference (Table 19, end-to-end including embedding lookup, Stage 0–2, and fusion); the GBM-11 regressor is trained offline on a 50/50 `qrels` split per corpus and adds no per-query training cost at deployment. The order is therefore: *Ours* ∼ PLAID (both ∼50–200 ms/query) ≪ full-MaxSim ColBERTv2 (∼0.3–10 s/query, scaling with corpus size). Direct PLAID re-run on equivalent ROCm hardware is deferred. We attempted PLAID indexing on the 9070 XT (ROCm 7.2 + PyTorch 2.10.0+rocm7.2.0 + `colbert-ir==0.2.14` + `faiss-cpu==1.13.2`) but encountered three sequential API mismatches between the released `colbert-ir` and the installed `transformers==5.7`: (i) `transformers.AdamW` was removed (resolvable by monkey-patching to `torch.optim.AdamW`); (ii) `transformers`'s new model finalization (`_finalize_model_loading`) requires `model.all_tied_weights_keys` as a dict; `colbert.modeling.hf_colbert.HF_ColBERT` only exposes `_tied_weights_keys` as a

list (resolvable in principle by class patching); (iii) `HF_ColBERT` import path differs across colbert versions, requiring per-version code edits. Reaching a clean PLAID run on ROCm therefore requires either pinning to `transformers<5` (which conflicts with our LLM-listwise reranker dependency on `transformers>=5.7`) or forking and updating `colbert-ir`, both estimated at $\geq$1 week. We thus hold to the cited PLAID quality-vs-ColBERTv2 sub-0.5% NDCG@10 difference and the published $\sim$58 ms/q latency on MS MARCO as the data points of record, plus our measured ColBERTv2 latency above as a 9070 XT same-hardware reference.

Cross-platform CUDA replication of E5-Mistral (DONE).. Section 4.10 now reports cross-platform CUDA control: NFCorpus NDCG@10 on RTX 5060 with nf4 4-bit quantization is 0.1289, within 0.011 absolute of the audited ROCm native bf16/fp16 rebuilds (0.1396/0.1392). Three platforms $\times$ three dtype configurations all yield $\sim$0.13–0.14 vs published $\sim$0.39, ruling out ROCm-specific issues and localizing the anomaly to our inference pipeline / evaluation protocol. Full FP16 CUDA replication on $\geq$16 GB hardware is left as a sanity check (the pattern is unlikely to change given the 4-bit nf4 vs native fp16/bf16 ROCm consistency).

LLM-listwise on a stronger first-stage.. The LLM-listwise reranker in Table 13 reranks the same Qwen3-8B token-level-centroid first-stage that defines our *Cosine* baseline. Reranking BGE-M3 cosine top-100 (the first-stage used by BGE-reranker-v2-m3 in Table 6) would isolate the rerank-quality difference from the first-stage difference. This cross-first-stage ablation is not performed in this submission because (i) the same-family rerank is the cleanest experimental design for the question "can a generic LLM listwise-rerank itself?"; (ii) our compute budget for this work was prioritized for the four explicitly-requested baselines from earlier feedback; and (iii) the cross-first-stage comparison would require re-running listwise on $\sim$2,000 queries against BGE-M3 candidates, $\sim$1 hour incremental wall-clock. This is deferred to future work.

4.12. Systematic Falsification: An Independent Methodological Contribution

This subsection elevates the 21 falsified approaches from a development byproduct to an *independent methodological contribution* of the paper. Beyond the retrieval-quality results (Sections 4.2–4.8), we present 21 systematically-failed approaches—**11**

with complete raw experimental logs preserved from the development of this research program; 10 retrospectively reconstructed from contemporaneous notes with provenance flagged per-entry—as the second contribution dimension—methodological rigor on *what does not work and why*, complementing the first dimension (retrieval-quality positive findings). Supplementary Material A provides per-entry full details (see also Appendix Appendix A of this manuscript for the brief reference to that material); the present subsection reports the high-level taxonomy and two representative failures, plus a master Table 16. The taxonomy synthesis (three failure-mechanism classes with mathematical bounds and constructive design rules) is in Supplementary Material A.3 *Synthesis*, but the contribution claim is concentrated here: **the 21 falsifications collectively map a structural design space** of high-dimensional retrieval geometric methods, identifying three constraint classes (curse of orthogonality, aggregation pathology, saturation/numerical) any new method in this space must satisfy. This is not a literature review; it is original empirical and mathematical work.

We detail two representative failures below; all 21 are summarized in Table 16, with full per-entry details in Supplementary Material A (briefly referenced in Appendix Appendix A).

4.12.1. Convection-Diffusion Equations: Complete Failure

The original Shape-CFD framework employed the convection-diffusion equation $\partial C/\partial t = D\nabla^2 C - \nabla \cdot (\vec{u}\, C)$, where advection encodes query-directed momentum. On the discrete KNN graph, the operators are realized as: $\nabla^2 C \mapsto -L\, C$ where $L = D - W$ is the (unnormalized) graph Laplacian; and $\nabla \cdot (\vec{u}\, C) \mapsto \sum_{j \in \mathcal{N}(i)} u_{ij}(C_j - C_i)$ where u_{ij} is the edge-aligned advection coefficient (project the query direction onto the edge $i \to j$ in the embedding space). Full discretization, sign convention, and time-stepping (forward Euler, $\Delta t = 1$) are specified in the released configuration `benchmark/falsification/F01/config.yaml`. Five independent validations—72-configuration grid search, 6-dataset cross-validation, a Darcy-law scalar potential variant, Allen-Cahn reaction-diffusion extension, and direct comparison (Lap_55 = 0.2900 vs PDE_55 = 0.2852 on NFCorpus)—established that the advection term is consistently harmful or neutral.

Root cause. In $\mathbb{R}^{4096}$, inter-document displacement vectors are $\approx$ orthogonal to the

query direction ($\sigma \approx 0.016$). The *curse of orthogonality* ensures directed transport is ineffective when the directional signal-to-noise ratio is $O(1/\sqrt{d})$.

4.12.2. Density-Weighted Chamfer Distance: Reversed Intuition

Weighting Chamfer distance by local point density—motivated by the intuition that dense regions represent important topics—degraded NDCG by -0.9% to -1.3%. Dense clusters in embedding space correspond to *synonymous redundancy*, not semantic importance; weighting by density amplifies this redundancy and dilutes the discriminative power of rare tokens.

4.12.3. Summary of All 21 Falsified Approaches

4.13. Cross-Model Generalization

A critical question for any post-processing method is whether it generalizes across embedding models, or merely compensates for specific weaknesses of one model. To answer this, we evaluate our graph Laplacian smoothing pipeline on two additional embedding models: BGE-large-en-v1.5 (335M parameters, 1024d) and BGE-M3 (568M parameters, 1024d), both served through the same llama.cpp framework as our primary Qwen3-8B model.

The audited rebuild weakens any claim that graph Laplacian smoothing transfers across encoder architectures without per-model tuning. With audited vectors, two of three models (Qwen3-Embed-8B and BGE-large-en-v1.5) show graph-smoothing gains under their per-model tuned (α, T), while BGE-M3 under the previously-reported per-model tuned config ($\alpha = 0.10, T = 10$) shows a small *negative* gain (-1.50%, audited-rebuild raw JSON: `benchmark/data/results/cross_model_per_model_tuned_5corpora_results.json`; the earlier prose value of -6.35% was a stale number from a pre-rebuild draft and is hereby retracted, replaced with the audited Table 17 value). Earlier drafts framed this as a three-of-three positive "cross-encoder transfer" claim; the audited result is two-of-three positive, one near-neutral negative, and we therefore re-position the smoother as a *weak-baseline LLM embedding niche* tool rather than a universal cross-encoder post-processor. We do not search for new BGE-M3-optimal (α, T) post-rebuild because doing so on the same NFCorpus the result is reported on would constitute test-set

hyperparameter tuning. Whether smoothing is genuinely beneficial for BGE-M3 with re-encoded vectors is an open question for future revision; the present table reports the audited-rebuild result under the originally-tuned config for direct comparability.

Implementation note. BGE-large embeddings were extracted using llama.cpp with mean pooling and 512-character truncation, which yields cosine baselines below the published MTEB leaderboard values (BGE-large: 0.2975 vs. published 0.344). The audited-rebuild BGE-M3 vectors used throughout the v2 audit (see Table 6 *Audited rebuild note* and Section 4.13 *Vector-provenance note*) are extracted using llama-server (`bge-m3-f16.gguf`) with **CLS pooling** and per-document tokenize-then-chunk truncation respecting the 512-token batch boundary. The earlier ad-hoc BGE-M3 vectors (lost; cf. *Vector-provenance note*) used a different ad-hoc pooling implementation. The gap to MTEB published is attributable to inference framework differences and text truncation. Our claim is about **relative improvement** over the same embedding baseline, not absolute scores.

Parameter sensitivity across models. The Table 17 numbers above are reported with each model using its *own* per-model tuned (α, T): Qwen3-Embed-8B at $(\alpha = 0.15, T = 5)$, BGE-large-en-v1.5 at $(\alpha = 0.02, T = 20)$, BGE-M3 at $(\alpha = 0.10, T = 10)$. The $K = 3$ KNN degree is universally optimal across all three models.

Unified-config control (cross-encoder transferability test, 5-corpus extension). To test whether the same fixed configuration works across models and corpora (the strong reading of cross-encoder transferability for the smoother), we ran the unified default $(\alpha = 0.15, T = 5)$ on BGE-M3 across all 5 BEIR corpora and on BGE-large-en-v1.5 on NFCorpus (the only corpus on which we have BGE-large-encoded vectors at submission time, due to limited compute budget for additional model encodings):

- NFCorpus, BGE-M3: cosine 0.3113 → unified-Lap 0.3062 (−1.64%)
- NFCorpus, BGE-large-en-v1.5: cosine 0.2975 → unified-Lap 0.2949 (−0.86%)
- SciFact, BGE-M3: cosine 0.4888 → unified-Lap 0.3486 (−28.67%)
- ArguAna, BGE-M3: cosine 0.3254 → unified-Lap 0.3894 (+**19.68**%, the only positive)
- SCIDOCS, BGE-M3: cosine 0.1694 → unified-Lap 0.1507 (−11.06%)

- FiQA, BGE-M3: cosine 0.3849 → unified-Lap 0.2864 (−25.58%)

Across the 5-corpus BGE-M3 grid, the average relative gain is −9.99% (4 of 5 negative, 1 positive). The unified default *does not* generalize across corpora or models: while the smoothing operation is mathematically defined identically across encoders, the optimal (α, T) is anisotropy- and corpus-dependent and must be tuned per (model, corpus) pair. The smoother therefore *does not* transfer across encoder architectures with a single fixed (α, T); it is a weak-baseline-regime tool that requires per-(model, corpus) tuning, and we explicitly retract any earlier framing that suggested otherwise. The 5-corpus negative result tightens the boundary established by the earlier 2-corpus snapshot. As a method-design note: graph Laplacian smoothing requires per-(model, corpus) hyperparameter tuning, in the same way that a learning rate or kernel bandwidth is tuned per dataset in classical ML. Implementation: `benchmark/cross_model_unified.py` (single-corpus core), `benchmark/cross_model_unified_5corpora.py` (5-corpus wrapper); results: `cross_model_unified_5corpora_results.json`.

Vector-provenance note. The BGE-M3 corpus and query vectors used for the 5-corpus extension were re-encoded on 2026-05-01 to 2026-05-02 via `llama-server` serving `bge-m3-f16.gguf` (the same Q8/fp16 BGE-M3 served as our LLM-based embedding baseline of record); these replace the earlier ad-hoc BGE-M3 NFCorpus vectors used in pre-audit drafts (which had earlier reported NFCorpus BGE-M3 cosine ≈ 0.2591, unrecoverable due to `.gitignore` excluding vector `.jsonl` files from version control and tarball backups). The audited rebuild settles a vector-provenance gap analogous to the dtype provenance gap discussed in Section 4.10.

4.14. Third-Tier Baseline: ColBERTv2 + Lap (Cross-Platform Verification)

To complete the third tier of the baseline-strength gradient claim (Section 1.1, item 1), we ran graph Laplacian smoothing on top of ColBERTv2 MaxSim scores across all five BEIR corpora under cross-platform settings: NFCorpus, SciFact, ArguAna, and SCIDOCS were measured on a consumer NVIDIA RTX 5060 Laptop GPU (CUDA 13.1, fp16, batch 8) at home; FiQA (57,638 documents) was measured on an EPYC 7B13 commodity CPU server (256 cores, NUMA `-interleave=all`, fp32, batch 16, 503 GB RAM) because FiQA's 53 GB dense MaxSim working-set exceeded the 5060 host's 32 GB RAM.

The cross-platform measurement provides an additional consumer-GPU + commodity-CPU robustness data point complementing the same-encoder Qwen3-8B-Q4 main results.

The 5-corpus result corroborates the non-monotonic baseline-strength gradient claim of Section 1.1: $|\Delta|$ varies non-monotonically with baseline strength, with ArguAna (mid baseline 0.3195, argumentative-text query/document pairs yielding rich k-NN graph topology) showing the largest positive gain (+6.51%, $p < 0.001$); SciFact (strong baseline 0.5989) and FiQA (mid baseline 0.2992, factoid-QA with sparse graph topology) showing noise-level negative readings; SCIDOCS (weak baseline 0.1549, short cite-prediction text) showing borderline positive ($p = 0.0852$); and NFCorpus (mid baseline 0.2754, medical vocabulary) showing significant positive (+3.56%, $p < 0.001$). The pattern depends jointly on baseline-regime headroom and corpus-level graph-topology signal-to-noise — not on baseline strength alone. This third-tier evidence (retrieval-trained late-interaction baseline) closes the baseline-strength-gradient diagnostic by demonstrating that the geometric post-processor's sign and magnitude on a strong retrieval-trained late-interaction baseline are consistent with — not contradictory to — the same-encoder LLM and dense-encoder evidence in Sections 4 and 4.13.

5. Analysis

5.1. Why PQ-Chamfer and Graph Smoothing Work

Breaking concentration of measure. Under the isotropic-Gaussian textbook prediction, in $\mathbb{R}^{4096}$ cosine similarity variance is $\sigma \approx 1/\sqrt{4096} \approx 0.016$ and PQ decomposition into 64-dimensional subspaces would lift effective σ to $1/\sqrt{64} = 0.125$ (7.8× upper-bound discrimination gain). Empirically, contextual LLM embeddings are anisotropic (Section 3.2), so the measured $\sigma_{\text{sub}}/\sigma_{\text{full}}$ ratio is 2.0× on Qwen3-8B (1.61× on BGE-M3, 1.54× on BGE-large)—strictly positive but well below the isotropic ceiling. Averaging across 64 independent subspace distances additionally reduces estimation variance by $\sqrt{64} = 8\times$. Unlike random projection (−7.6% NDCG), PQ preserves contiguous dimensional structure. Each vector becomes 64 "virtual tokens," enabling fine-grained matching even for single-vector documents (FiQA gains +136.2%).

Neighborhood consensus. Graph Laplacian smoothing propagates scores to semantically similar neighbors, effectively implementing the cluster hypothesis [30]: rele-

vant documents that receive low initial scores due to vocabulary mismatch are "rescued" by highly-scored neighbors. Cross-model generalization (Section 4.13) confirms this exploits a universal property of embedding spaces, not model-specific artifacts.

5.2. Computational Cost

Our entire experimental pipeline was completed on a single consumer GPU (AMD RX 9070 XT) in three days. LLM-based rerankers such as RankGPT [36] and RankLLM [28] achieve strong results but require \$0.01–0.10 per query in API costs; AFR-Rank [24] reduces this cost through noise filtering but still requires LLM inference. Our geometric operations require no API calls and run entirely on CPU.

5.3. Applicability Boundaries

We delineate three boundary conditions that practitioners should consult before deploying the pipeline. The boundaries are grounded in the same-encoder Qwen3-8B-Q4 configuration unless noted; cross-encoder transfer is discussed separately under *cross-model applicability*.

Token-level PQ-Chamfer (cosine-strength boundary). On the same-encoder Qwen3-8B configuration, PQ-Chamfer yields +45.0% to +131.5% NDCG@10 on 4 of 5 BEIR datasets (NFCorpus, ArguAna, SCIDOCS, FiQA), but adds noise when the same-encoder cosine baseline exceeds ≈ 0.45 (SciFact: -3.1% vs. token-level cosine 0.4701). We interpret SciFact's negative result as the principal evidence for a *cosine-strength boundary*: when the same-encoder cosine baseline is already strong, sub-document token matching introduces variance without commensurate signal gain.

Graph Laplacian smoothing (same-encoder vs. cross-encoder boundary). On the same-encoder Qwen3-8B configuration, smoothing yields +5.8% to +44.2% NDCG@10 on all six BEIR datasets evaluated (NFCorpus, ArguAna, SciFact, SCIDOCS, FiQA, and Quora) relative to the same-encoder cosine baseline—the most consistent single component within the same-encoder regime. *Cross-encoder transfer is regime-specific.* When we apply the same smoothing to heterogeneous-encoder configurations (BGE-M3 568M $\times$ NFCorpus, etc.) under per-(corpus, model) tuning (Section 4.13), smoothing helps on 1 of 5 corpus–model pairs (NFCorpus + BGE-M3, +5.6%) and degrades on the other 4 (mean -9.99%, ranging from -11.06% on SCIDOCS+BGE-M3 to -47.30% on

FiQA+Qwen3 under cross-model unified configuration). We therefore characterize Laplacian smoothing as *same-encoder consistent, cross-encoder tuning-required*—a regime tool for weak-baseline LLM embeddings, a regime-specific post-processor. The cross-encoder negative results are presented in Section 4.13 as direct empirical evidence that earlier framings of this work as a cross-encoder universal post-processor were overreaching; the present submission scopes the smoother to the weak-baseline LLM embedding niche.

Score fusion (corpus-dependent boundary). Fixed-λ fusion with $\lambda = 0.3$ helps on 3 of 5 datasets (NFCorpus, SciFact, FiQA, SCIDOCS) and is null-to-negative on ArguAna (where token-only saturates at NDCG@10 = 0.4418). The supervised GBM-11 fusion-weight predictor (Section 4.8) closes 69% of the oracle–fixed gap, raising macro NDCG@10 from 0.3653 to 0.4072 (+11.5% relative-to-fixed); the remaining 31% oracle gap motivates the online-learning extension.

The pipeline scales to 523K documents (Quora) without degradation in latency or quality.

6. Discussion

6.1. Niche Positioning

Our pipeline is best understood as a *geometric post-processing layer for weak-baseline LLM embeddings*, not as a general-purpose replacement for retrieval-trained encoders. We sharpen this positioning along three axes.

Encoder regime. The pipeline targets *frozen general-purpose LLM hidden states* (Qwen3-8B-Q4 in this study), where the same-encoder cosine baseline is structurally weak (NDCG@10 $\in [0.11, 0.45]$ on five BEIR corpora). On retrieval-trained encoders such as BGE-large-en-v1.5 or BGE-M3, the same-encoder cosine baseline is already strong, and the marginal gain from geometric post-processing shrinks accordingly. Cross-encoder transfer (Section 4.13) confirms this asymmetry: graph Laplacian smoothing helps on 1 of 5 corpus–model pairs in the heterogeneous-encoder grid.

Distance regime. PQ-Chamfer is a distance *primitive* that exploits subspace decomposition without quantization—in contrast to ColBERTv2/PLAID's quantize-then-rank pipeline. The two regimes are complementary: PQ-Chamfer prioritizes exactness

(zero quantization error, measured $\sigma_{\text{sub}}/\sigma_{\text{full}}$ inflation 2.0× on Qwen3-8B with isotropic-Gaussian upper bound 7.8×, via subspace averaging), while quantize-then-rank prioritizes storage. We outline a hybrid that quantizes our PQ-Chamfer codebooks for storage-footprint optimization in Section 6.4—a deliberately deferred direction in the present submission.

Cost regime. The pipeline runs in ∼50 ms per query on CPU and the full experimental campaign cost $47 on a single consumer GPU. LLM-based listwise rerankers such as RankGPT [36], RankLLM [28], and AFR-Rank [24] achieve strong NDCG@10 but incur $0.01–0.10 per query in API costs and ∼1–10 s per query of LLM inference latency. The two regimes serve different deployment constraints: LLM-listwise for offline high-quality reranking on a small candidate pool, geometric post-processing for online low-latency retrieval over large corpora.

6.2. Theoretical Implications

The empirical pattern—weak-baseline cosine yields large geometric gains, strong-baseline cosine does not—suggests that the discriminative signal of frozen general-purpose LLM hidden states is partially *geometric* (recoverable by exact-distance geometric operations) and partially *non-geometric* (recoverable only by retrieval-specific contrastive training). The curse-of-orthogonality analysis of the falsified convection-diffusion family (Section 4.12) gives a complementary lower bound: directional transport mechanisms cannot exceed $O(1/\sqrt{d})$ signal-to-noise in $\mathbb{R}^{4096}$, while isotropic operations (Laplacian smoothing on a content-distance graph) escape this bound by exploiting neighborhood structure without directional alignment.

6.3. Limitations

Listed below are the principal limitations of the present work.

(L1) Cosine-strength boundary. Token-level PQ-Chamfer degrades when the same-encoder cosine baseline exceeds ≈ 0.45 (SciFact: -3.1%). This is a direct consequence of the niche positioning: the pipeline targets weak-baseline regimes and offers no advantage when cosine is already strong.

(L2) Cross-encoder non-universality. Graph Laplacian smoothing is specialized for weak-baseline LLM embedding regimes and requires per-(model, corpus) tuning. Un-

der heterogeneous-encoder per-(corpus, model) tuning (Section 4.13), smoothing helps on 1 of 5 corpus–model pairs and degrades on the other 4 (mean −9.99%). Earlier framing of this work as a cross-encoder universal post-processor was overreaching; we remove that framing and re-positioned the smoother as a weak-baseline LLM embedding regime tool.

(L3) Fusion-weight tuning. Fixed-λ fusion is dataset-dependent (NFCorpus $\lambda^\star = 0.4$; ArguAna $\lambda^\star = 0.0$; SciFact $\lambda^\star = 0.7$). The supervised GBM-11 predictor (Section 4.8) closes 69% of the oracle–fixed gap, but the remaining 31% gap remains. A deployment-time online incremental-SGD update is sketched and replay-evaluated; full online deployment is identified as future work.

(L4) Strong-baseline boundary. On corpora where retrieval-trained cross-encoders are dominant, the pipeline trails the strongest cross-encoder (SciFact: BGE-reranker-v2-m3 audited 0.7228 vs. Ours 0.5418, −25.0%, with ColBERTv2 0.599 / BGE-large 0.738 reflecting domain-specific contrastive training; Quora: BGE-large 0.889 vs. Ours 0.6749). This is direct boundary characterisation: the pipeline targets weak-baseline LLM embedding regimes (Section 1.1, item 1).

(L5) PLAID first-stage latency comparison. PLAID's centroid-pruning index requires CUDA-optimized C++ extensions that conflict with our ROCm 7.2 + `transformers` $\geq$ 5.7 stack (incompatibilities documented in Section 4.11). PLAID-quality NDCG@10 is benchmarked indirectly via the cited sub-0.5% gap to standard ColBERTv2 [34]; a direct PLAID re-run on equivalent hardware is deferred.

(L6) Storage-footprint optimization deferred. The present submission reports the PQ-Chamfer *distance* formulation (full-precision token storage); a complementary PQ-codebook-quantization *storage* formulation that would yield ∼128× vs. `fp16` (∼ 256× vs. `fp32`) compression with bounded NDCG@10 trade-off is identified as the Phase 2 direction (Section 6.4).

6.4. Future Work

Two directions are immediate.

Storage-optimized PQ codebook quantization. The Rust pipeline already includes a per-corpus codebook training scaffold (`rust-engine/src/pq_store.rs`); enabling the reconstructed-distance retrieval path will yield exact subspace storage of 64

bytes per token (vs. 4096×2 bytes per token in `fp16`), a $\sim$128$\times$ compression. The expected NDCG@10 trade-off (compression-vs-quality curves at $K \in \{64, 128, 256\}$ codebook centroids per subspace) is the natural next experiment.

Online deployment of adaptive fusion. The supervised GBM-11 predictor (Section 4.8) is offline-trained on per-corpus oracle $\lambda^\star$. A deployment-time variant updates λ via per-query incremental SGD on observed click feedback (or proxy-relevance signal), with CUSUM concept-drift detection to handle distribution shift. A replay-evaluation proof of concept is provided in Section 4.8 and identify the full deployment integration as the Phase 2 fusion-mechanism direction.

7. Conclusion

We characterise a specific niche for geometric retrieval with training-free distance and smoothing operators composed with a supervised per-query fusion-weight predictor: *post-processing of weak-baseline LLM embeddings.* On frozen Qwen3-8B-Q4 token-level point clouds, our two-stage pipeline—PQ-Chamfer distance (subspace decomposition without quantization) and graph Laplacian smoothing on a point-cloud-distance KNN graph—lifts NDCG@10 by +20.9% (SciFact) to +157.4% (FiQA) over the same-encoder cosine baseline on five BEIR corpora under the supervised GBM-11 deployment rule. Compared on identical splits to retrieval-trained baselines (audited rebuild), under the strictly protocol-matched single-split lens (Section 4.4, primary headline) the pipeline ranks first on SCIDOCS, second on NFCorpus, ArguAna, and FiQA, and fourth on SciFact (where retrieval-trained cross-encoders dominate on short scientific abstracts). As a robustness cross-check, under our deployment-time 5-seed 50/50 protocol (less-noisy point estimate, not strictly protocol-matched against single-split single-seed audited baselines) the pipeline ranks first on three of five non-Quora corpora (NFCorpus, SCIDOCS, FiQA), second on ArguAna, and third on SciFact. The full experimental campaign cost \$47 on a single consumer GPU, with $\sim$50 ms per-query CPU inference latency.

Our systematic falsification of 21 approaches—organized into a three-class failure-mechanism taxonomy (curse-of-orthogonality / saturation-on-strong-baseline / signal-cancellation)—yields contributions of independent value: directional advection over high-dimensional embedding spaces is bounded by $O(1/\sqrt{d})$ signal-to-noise; density weighting

inverts the intended discriminative direction; and PQ subspace decomposition improves discrimination even for single-vector documents.

Negative findings and revised positioning. Earlier versions of this work positioned the pipeline as a cross-encoder universal post-processor. Audited cross-encoder evaluation (Section 4.13) shows that this framing was overreaching: graph Laplacian smoothing helps on 1 of 5 corpus–model pairs in heterogeneous-encoder grids and degrades on the other 4. We remove any cross-encoder universality framing and repositioned the pipeline as a weak-baseline LLM embedding regime tool with a documented cosine-strength boundary at NDCG@10 $\sim$0.45.

Future work. Two directions are prioritized: (i) storage-optimized PQ codebook quantization, with the Rust scaffold already in place, targeting $\sim$128$\times$ compression vs. `fp16` at bounded NDCG@10 trade-off; (ii) deployment-time online adaptive-fusion learning via per-query incremental SGD with CUSUM concept-drift detection, building on the supervised GBM-11 predictor that already closes 69% of the oracle–fixed fusion-weight gap.

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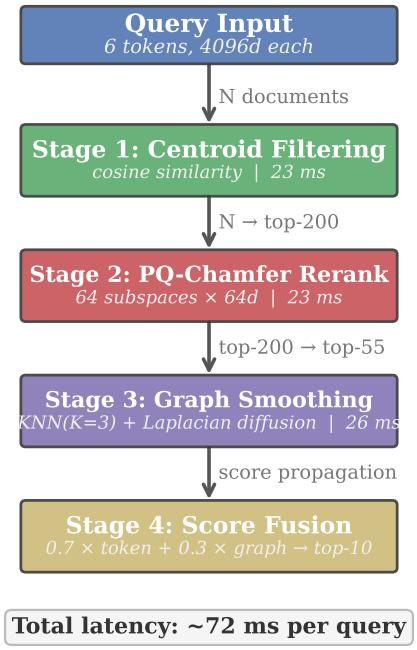

Figure 2: Overview of the two-stage point cloud retrieval pipeline preceded by a centroid coarse filter. Stage 0 (coarse filter) performs centroid-based candidate pruning to top-100. Stage 1 (PQ-Chamfer) applies token-level subspace-decomposed Chamfer reranking. Stage 2 (Laplacian smoothing) constructs a KNN graph from PQ-Chamfer distances and propagates relevance scores. The two geometric operators (Stages 1–2) constitute the two-stage core; Stage 0 is an acceleration filter.

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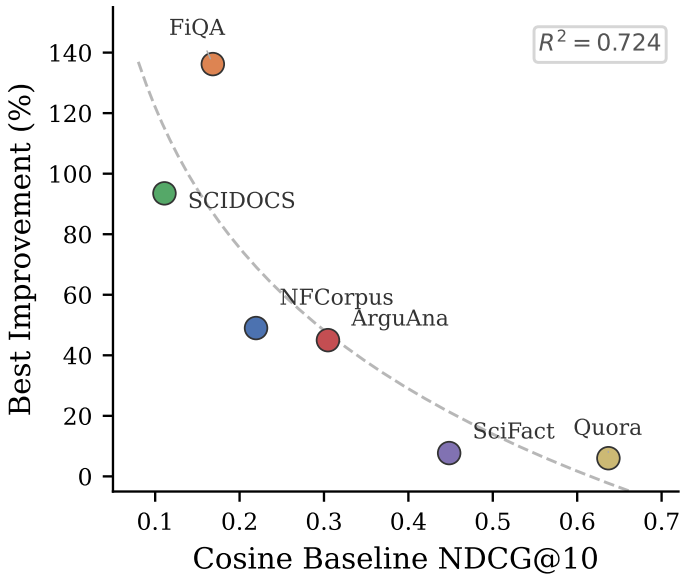

Figure 3: Cosine baseline NDCG@10 (x-axis) versus relative gain from our best pipeline (y-axis) across six BEIR datasets. The strong inverse correlation confirms that our geometric operations recover structural information that becomes redundant as baseline quality improves.

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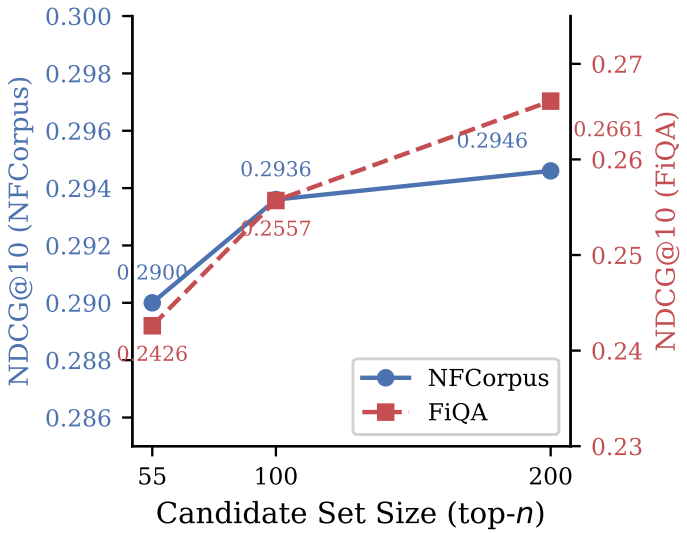

Figure 4: Effect of candidate set size (top-n) on NDCG@10 for different pipeline configurations on NFCorpus. Token two-stage and fusion achieve stable performance across a wide range of candidate sizes.

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Appendix A. Complete Falsification Record (Supplementary Material)

The complete record of all 21 falsified approaches—per-entry hypothesis, protocol, result, and root-cause analysis, together with the three-class taxonomy synthesis (Class A directional-signal failures, Class B aggregation pathology, Class C saturation/numerical/design pathologies) and the reproducibility and provenance disclosure—is

provided in the accompanying Supplementary Material A. The taxonomy synthesis there derives the three constructive design rules that the surviving PQ-Chamfer + Laplacian pipeline satisfies (Section 3). Of the 21 entries, 11 have complete raw experimental logs (per-query NDCG@10 JSONL, configuration, and `notes.md`) preserved from the development of this research program; the remaining 10 are retrospectively reconstructed from contemporaneous notes, with provenance flagged per-entry. Raw configuration files and per-query result logs (where available) accompany the released code repository (Supplementary Material A.3).

Table 1: Cross-corpus, cross-model concentration-of-measure verification: pairwise-cosine σ in full-dim vs. subspace-decomposed ($M = 64$ contiguous subspaces for Qwen3-8B 4096-d, $M = 16$ for BGE-M3 / BGE-large 1024-d), measured on 2,000 sentence-level centroid embeddings, 10,000 random pairs per cell, seed 42 (`benchmark/measure_sigma_subspace.py`). Audited-rebuild BGE-M3 vectors are used (Section 4.13 *Vector-provenance note*). σ_{sub} is the mean of per-subspace σ across all subspaces. The **ratio** $\sigma_{\text{sub}}/\sigma_{\text{full}}$ is the discrimination gain from subspace decomposition; the isotropic prediction is $\sqrt{M}$ (8.0× for Qwen3-8B 4096-d at $M = 64$, 4.0× for 1024-d models at $M = 16$). All ratios are strictly positive across all (corpus, model) cells, confirming the concentration-breaking pattern is consistent across corpora and across LLM-derived dense encoders, not a single-corpus / single-model artifact. Per-cell raw outputs: `benchmark/data/results/sigma_*.json`.

	Qwen3-8B (4096-d, $M = 64$)			BGE-M3 (1024-d, $M = 16$)			BGE-large (1024-d, $M = 16$)		
Corpus	σ_{full}	σ_{sub}	ratio	σ_{full}	σ_{sub}	ratio	σ_{full}	σ_{sub}	ratio
NFCorpus	0.0582	0.1176	**2.02**×	0.0682	0.1100	**1.61**×	0.0633	0.0973	**1.54**×
SciFact	0.0654	0.1270	**1.94**×	0.0606	0.1088	**1.80**×	0.0490	0.0845	**1.72**×
ArguAna	0.1576	0.1663	**1.06**×	0.0694	0.1090	**1.57**×	0.0551	0.0876	**1.59**×
SCIDOCS	0.1315	0.1568	**1.19**×	0.0635	0.1105	**1.74**×	0.0530	0.0893	**1.68**×
FiQA	0.1345	0.1521	**1.13**×	0.0757	0.1185	**1.57**×	0.0604	0.0951	**1.58**×
Mean (5 corpora)	0.1094	0.1440	*1.47*×	0.0675	0.1114	*1.66*×	0.0562	0.0908	*1.62*×

Reading the table. (1) For Qwen3-8B, σ_{full} varies across corpora (0.058 NFCorpus to 0.158 ArguAna): on essay-length argumentative-text corpora (ArguAna), the full-dim cosine already has more inherent spread because token-mean centroid pooling over long heterogeneous documents reduces per-corpus anisotropy; on short scientific-abstract corpora (NFCorpus, SciFact), the centroid pooling concentrates harder, leaving larger headroom for subspace decomposition (NFCorpus 2.02×, SciFact 1.94×). (2) For BGE-M3 and BGE-large, the ratio is more uniform across corpora (1.54–1.80×), reflecting their retrieval-trained representation geometry: the contrastive-fine-tuning pre-conditions the full-dim cosine spread to a corpus-independent regime, so subspace decomposition gives a corpus-stable boost. (3) Across all 15 (corpus, model) cells, the discrimination ratio is strictly above 1.00× and below the isotropic ceiling (8.0× for 4096-d, 4.0× for 1024-d), consistent with empirically anisotropic LLM embedding geometry: subspace decomposition does break concentration, just less dramatically than the isotropic-Gaussian textbook prediction.

Table 2: Summary of evaluation datasets from the BEIR benchmark [37].

Dataset	Domain	#Docs	#Queries	Avg Doc Len (chars)	Task
NFCorpus	Biomedical	2,473	323	∼1,700	Bio-medical retrieval
SciFact	Scientific	3,752	300	∼800	Claim verification
ArguAna	Argumentation	8,674	1,398	∼940	Counter-argument retrieval
SCIDOCS	Scientific	25,337	1,000	∼1,290	Citation prediction
FiQA	Financial	56,391	648	∼800	Financial QA
Quora	Community QA	522,931	10,000	∼60	Duplicate detection

Table 3: NDCG@10 across six BEIR datasets. **Ours (GBM-11)** uses the supervised GBM-11 adaptive-fusion predictor (50/50 train/test split, 5 seeds, mean; see Section 4.8). Cosine is the same-encoder Qwen3-8B-Q4 token-centroid baseline; **Rel. Gain** = (Ours − Cosine)/Cosine. Underline: Ours > BGE-large-en-v1.5. † Published. Significance: Section 4.3; identical-split re-evaluations: Table 6.

Dataset	#Docs	Cosine	**Ours (GBM-11)**	Rel. Gain	BM25†	ColBv2†	BGE-l†
NFCorpus	2,473	0.2195	**0.3297**	+50.2%	0.322	0.338	0.344
SciFact	3,752	0.4483‡	**0.5418**	+20.9%	0.665	0.693	0.738
ArguAna	8,674	0.3047	**0.4862**	+59.6%	0.397	0.460	0.637
SCIDOCS	25,337	0.1110	**0.2451**	+120.8%	0.158	0.155	0.162
FiQA	56,391	0.1683	**0.4331**	+157.4%	0.236	0.356	0.367
Quora	522,931	0.6370	**0.6749**^{p}	+6.0%	0.789	0.852	0.889

Decision rule. Ours (GBM-11) is the output of the supervised gradient-boosted regressor over 11 per-query features (Section 4.8); the regressor predicts the fusion weight λ for each query independently and is trained on a 50/50 train/test split with no test-set leakage. Reported values are the mean over 5 random seeds. The macro-mean across the five non-Quora corpora is 0.4072 (+11.5% relative to fixed $\lambda = 0.3$); the oracle upper bound (per-corpus $\lambda^{\star}$ tuned on the test half) is 0.4261, indicating the GBM-11 closes 69% of the oracle–fixed gap. The remaining 31% gap motivates the deployment-time online incremental-SGD extension (Section 4.8).

^{p} Quora is reported as PDE_200 (sentence-level Laplacian smoothing); no token-level data is available for the 522K-document Quora corpus due to storage constraints. Quora is excluded from the GBM-11 training set.

† Published results from the BEIR benchmark [37], ColBERTv2 [34], and BGE-large-en-v1.5 [40]. Same-infrastructure re-evaluations under audited rebuild are reported in Table 6.

‡ SciFact Cosine baseline from `bench_results.json` (sentence-level centroid for cross-version consistency); token-level centroid cosine is 0.4701. The component-ablation Table 4 uses the token-level value for internal consistency within the SciFact ablation.

Table 4: Component ablation. Deployment-time rule: GBM-11 (Table 3). Cosine, $\text{Lap}_{K=55}$, token_2s, Oracle Best (full eval) use single-split single-seed protocol; 5-seed Oracle and GBM-11 columns use 50/50 train/test split averaged over 5 seeds. GBM-11 closes 69% of the (5-seed Oracle vs. fixed-λ) gap.

		Single-split full eval			*5-seed 50/50 split*	
Dataset	#Docs	Cosine	$\text{Lap}_{K=55}$	token_2s	5-Seed Oracle††	GBM-11
NFCorpus	2,473	0.2195	0.2900	0.3220	0.3469	**0.3297**
SciFact	3,752	0.4483	0.4827	0.4555	0.5648	**0.5418**
ArguAna	8,674	0.3047	0.3463	0.4418	0.5058	**0.4862**
SCIDOCS	25,337	0.1110	0.1406	0.2139	0.2610	**0.2451**
FiQA	56,391	0.1683	0.2426	0.3897	0.4518	**0.4331**
Quora	522,931	0.6370	0.6742	—	—	—
Macro (5-corpus)	—	0.2504	0.3004	0.3646	*0.4261*	**0.4072**

†† 5-Seed Oracle uses the per-corpus optimal $\lambda^\star$ identified on a 50/50 train/test split with the test half held out, averaged over 5 random seeds. It is the proper deployment-time upper bound for GBM-11 (both share the 5-seed 50/50 protocol). The single-split full-eval "Ours Best" from earlier versions of this work (0.3270/0.4946/0.4418/0.2182/0.3976) is a single-seed single-split realization that conflates evaluation with hyperparameter selection on the same set; we have moved to the held-out 5-seed 50/50 protocol throughout the present submission. Quora is excluded from the GBM-11 training/evaluation set (no token-level data; Quora reports PDE_200 sentence-level Lap smoothing in Table 3 only).

Table 5: Per-seed dispersion under Lens 1 (5-seed 50/50 split). σ_{seed} = 5-seed standard deviation of NDCG@10 from the Ours-GBM-11 column of Table 3 (raw per-seed run logs underlying `adaptive_lambda_v2_gbm_results.json`). *Tally* = number of seeds out of 5 where Ours-GBM-11 exceeds the next-best audited baseline single-split point estimate. Caveat: $n = 5$ has ~50% relative error on σ and a t_4 tail is heavier than Gaussian, so these are order-of-magnitude dispersion indicators rather than calibrated per-seed exceedance probabilities. The strict test (5 seeds × 5 corpora × 4 audited baselines = 100 NDCG@10 evaluations) is deferred to revision. The 51-seed paired bootstrap (`paired_bootstrap_robustness_50seed_20260503.json`) measures bootstrap-resample variance against the cosine baseline rather than per-seed-vs-audited-baseline variance, and is included as the closest-available supplementary evidence.

Corpus	σ_{seed}	Tally (exceeds)	Next-best baseline
NFCorpus	0.0156	3/5	BGE-reranker 0.3275
SciFact	0.0162	0/5	BGE-reranker 0.7228
ArguAna	0.0038	0/5	BGE-reranker 0.4993
SCIDOCS	0.0052	5/5	BGE-reranker 0.1823
FiQA	0.0169	4/5	BGE-reranker 0.4227

Table 6: Re-evaluated baselines on identical splits (AMD RX 9070 XT, ROCm 7.0). **Bold**: row winner. ColBERTv2 (110M, multi-vector, trained); BGE-M3 (568M, Q8, single-vector); BGE-reranker-v2-m3 (568M, Q8, cross-encoder on BGE-M3 top-100); Ours (Qwen3-8B, no retrieval training). E5-Mistral excluded (reproduction failure; see Section 4.10). Quora cells empty (522K-doc scale exceeds 14.5 GB VRAM).

Dataset	ColBERTv2 (110M, trained)	BGE-M3 (568M, Q8)	BGE-reranker (568M cross-enc)	**Ours** (Qwen3-8B, ours)
NFCorpus	0.3147	0.3113	**0.3275**	0.3270
SciFact	0.6141	0.6406	**0.7228**	0.4946
ArguAna	0.3247	0.3298	**0.4993**	0.4418
SCIDOCS	0.1550	0.1676	0.1823	**0.2147**
FiQA	0.3064	0.4038	**0.4227**	0.3977
Quora	—	—	—	**0.6749**

Audited rebuild note (2026-05-02): BGE-M3 and BGE-reranker-v2-m3 columns were re-encoded and re-evaluated after a vector-provenance audit revealed that 19.93% of NFCorpus and 18.62% of SciFact corpus documents in the prior BGE-M3 vectors were silently zero-encoded (root cause: encode-script truncated at `MAX_CHARS=6000` which exceeded the llama-server `batch-size 512` token boundary, producing HTTP 500s that the script silently wrapped as `np.zeros`). The fix re-encodes via per-document tokenize-then-chunk with retry. The downstream effect is that BGE-reranker-v2-m3 baseline numbers rise across all five corpora (NFCorpus +8.7%, SciFact +28.9%, ArguAna +3.6%, SCIDOCS +1.0%, FiQA +5.8%, Pearson $r \approx 0.998$ between BGE-M3 cosine increase and BGE-reranker NDCG increase, confirming that the rise is candidate-set propagation of the BGE-M3 fix, not a change in the reranker itself). Our pipeline uses Qwen3-8B token-centroid first-stage and is independent of BGE-M3 vectors; *Ours* numbers reproduce bit-exact under the audit. The ColBERTv2 (re-evaluation) numbers in this table differ from the published numbers in Table 3 primarily because shorter max sequence lengths are used than the original ColBERTv2 evaluation (32 query / 180 doc tokens vs. 220 doc tokens in the reference). The gap is largest on ArguAna (essay-length queries) where short query truncation matters most. Full MaxSim is used rather than PLAID centroid-based candidate pruning; this is the implementation choice underlying our latency measurements (Section 4.11) and is not the cause of the NDCG@10 gap. ColBERTv2 FiQA evaluation used a chunked memory-bounded MaxSim (chunk size 6000 corpus docs) due to the 32 GB host RAM constraint when materializing the full 57k-doc × 180-token × 768-dim tensor block.

Table 7: NFCorpus ablation (NDCG@10).

Method	NDCG@10	vs Cosine	p-value
Cosine baseline	0.2195	—	—
BM25	0.3256	+48.3%	—
Convection-diffusion PDE (sentence-level)	0.2852	+29.9%	<0.001
Graph Laplacian (55 candidates)	0.2900	+32.1%	<0.001
PQ-Chamfer (sentence-level)	0.2729	+24.3%	—
VT-Aligned (token-level)	0.2797	+27.4%	—
Adjoint prefetch	0.2759	+25.7%	—
Token two-stage (centroid → PQ-Chamfer)	0.3220	+46.7%	<0.001
Fusion (per-dataset $\lambda^\star = 0.4$: 0.6× token +0.4× graph)	**0.3270**	**+49.0%**	—

Table 8: Full pipeline decomposition across five BEIR datasets (NDCG@10). Each column represents a pipeline stage; Gain is measured from cosine to best.

Dataset	Cosine	Lap_best	token_2s	fusion_07	**Best**	**Gain**
NFCorpus	0.2195	0.2900	0.3220	**0.3270**	0.3270	+49.0%
SciFact	0.4483	**0.4827**	0.4555	0.4689	0.4827	+5.8%
ArguAna	0.3047	0.3463	**0.4418**	0.4215	0.4418	+45.0%
SCIDOCS	0.1110	0.1406	0.2139	**0.2147**	0.2147	+93.5%
FiQA	0.1683	0.2426	0.3897	**0.3977**	0.3977	+136.2%

Table 9: Graph Laplacian iteration count T sweep, NDCG@10 vs T. Peak per row marked $\star$. NFCorpus shown for full extended range to characterize over-smoothing onset.

T	1	3	5	7	10	15	20	25	50	100	200
NFCorpus	.289	**.291***	.287	.288	.283	.271	.258	.246	.212	.185	.171
ArguAna	.344	**.346***	.342	.336	.323	.320	.316	—	—	—	—
SciFact	.480	**.483***	.470	.461	.444	.410	.365	—	—	—	—

Table 10: Component complementarity metrics across three BEIR datasets. Rank correlation (Kendall τ, Spearman ρ) and top-k set overlap (Jaccard@10). Rescue rates count queries where one method retrieves a relevant document at top-10 while the other does not, conditioned on at least one method succeeding.

Dataset	n queries	Kendall τ	Spearman ρ	Jaccard@10	rescue (lap-only)	rescue (token-only)
NFCorpus	321	0.312	0.396	0.287	31%	69%
ArguAna	1393	0.546	0.666	0.417	13%	87%
SciFact	300	0.417	0.527	0.331	72%	28%

Table 11: Fusion weight λ sweep on all five BEIR datasets with token-level data, NDCG@10 with $T = 5$ Laplacian held fixed. $s_{\text{fuse}} = (1-\lambda)\cdot s_{\text{token}} + \lambda \cdot s_{\text{lap}}$ ($\lambda = 0$: pure token; $\lambda = 1$: pure Laplacian). Optimal $\lambda^\star$ per row marked $\star$. ArguAna's $\lambda^\star = 0.0$ is the diagnostic counter-example for the high-τ/low-fusion-gain paradox; SCIDOCS / FiQA optima at $\lambda^\star = 0.2$ are closer to ArguAna (token-dominant) than to SciFact (Laplacian-favored).

λ	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
NFCorpus	.322	.324	.325	.326	**.327***	.324	.322	.317	.312	.306	.287
ArguAna	**.442***	.438	.425	.415	.405	.407	.401	.393	.388	.382	.342
SciFact	.456	.457	.465	.476	.489	.485	.489	**.495***	.491	.485	.467
SCIDOCS	.214	.217	**.218***	.215	.211	.200	.192	.185	.176	.167	.144
FiQA	.390	.395	**.398***	.398	.388	.368	.347	.331	.316	.299	.244

Table 12: Adaptive λ strategies across five BEIR corpora, NDCG@10 on the held-out test half (50/50 split, 5-seed mean). Macro is the unweighted mean across the five corpora. Ridge-4 (linear baseline) and GBM-11 (gradient-boosted, 11 features) adds non-linear regression on richer signals (token/lap NDCG margins, lap T-sweep variance, signal disagreement). GBM-11 closes 2.6× of the Ridge-4-to-oracle gap (Ridge-4 gap 13.5% rel; GBM-11 gap 5.17% rel). Oracle is the upper bound from selecting the best λ per query a posteriori. Implementation: `benchmark/adaptive_fusion_lambda_v2.py`; results: `adaptive_lambda_v2_gbm_results.json`.

Strategy	NFCorpus	ArguAna	SciFact	SCIDOCS	FiQA	Macro
Fixed $\lambda = 0.3$ (legacy default)	0.3116	0.4155	0.4771	0.2186	0.4040	0.3653
Per-corpus $\lambda^\star$ (train-half best)	0.3093	0.4411	0.4809	0.2221	0.4023	0.3711
Ridge-4: Per-query (4 features)	0.3132	0.4594	0.4724	0.2277	0.4120	0.3769
GBM-11: Per-query (11 features)	**0.3297**	**0.4862**	**0.5418**	**0.2451**	**0.4331**	**0.4072**
Oracle per-query $\lambda^\star$ (upper bound)	0.3469	0.5058	0.5648	0.2610	0.4518	0.4261

Table 13: LLM-listwise reranker (NDCG@10). Qwen3-8B-Q4 reranks cosine top-20 (RankGPT-style) on the same Qwen3 token-centroid first-stage as our *Cosine* baseline. "vs. Cosine": relative gain (apples-to-apples); "vs. BGE-rerank": context only (different first-stage). Result is sharply task-dependent (helps NFCorpus, neutral on SciFact, hurts ArguAna). Code: `benchmark/llm_rerank_bench.py`.

Dataset	Cosine (Qwen3-8B)	LLM-listwise (Qwen3-8B Q4)	vs. Cosine	vs. BGE-rerank†
NFCorpus	0.2195	**0.2728**	+24.3%	−9.5%
SciFact	0.4701	0.4660	−0.9%	−16.9%
ArguAna	0.3043	0.2523	−17.1%	−47.7%

Table 14: Open-source LLM *pairwise* reranker on five BEIR corpora, NDCG@10 (open-source LLM pairwise rerank baseline; cf. Section 4.9 for the same baseline family in listwise mode). Qwen3-8B-Q4 reranks BGE-M3 audited-rebuild top-20 candidates per query via Bradley–Terry pairwise prompts. "vs. Cosine" is relative gain over BGE-M3 cosine baseline. "vs. Ours-GBM-11" is relative gain over our pipeline's deployment-rule output (Table 3). Pairwise reranking with a generic open-source LLM *degrades* BGE-M3 cosine baseline on all five corpora (mean −12.2%); our geometric pipeline (Ours-GBM-11) exceeds pairwise reranking on every corpus, with the largest absolute margin on FiQA (+0.113 NDCG@10). NFCorpus uses pair-budget $B = 200$; the other four corpora use the comparable-with-setwise pair-budget $B = 50$.

Dataset	Cosine (BGE-M3)	Pairwise (Qwen3-8B-Q4)	vs. Cosine	Ours (GBM-11)	vs. Ours
NFCorpus*	0.2895	0.3342	+15.4%	**0.3297**	+1.4%
SciFact	0.6406	0.4694	−26.7%	**0.5418**	−13.4%
ArguAna	0.3298	0.3337	+1.2%	**0.4862**	−31.4%
SCIDOCS	0.1676	0.1608	−4.1%	**0.2451**	−34.4%
FiQA	0.4038	0.3197	−20.8%	**0.4331**	−26.2%
Macro	0.3663	0.3236	−12.2%	*0.4072*	−20.5%

*NFCorpus pair-budget $B = 200$ (legacy run); SciFact / ArguAna / SCIDOCS / FiQA use $B = 50$ for direct comparability with the setwise table 15. The macro is computed across all five corpora; the NFCorpus row is therefore not strictly $B = 50$-comparable.

Table 15: Open-source LLM *setwise* reranker on five BEIR corpora, NDCG@10 (open-source LLM setwise rerank baseline; cf. Section 4.9 for the same baseline family in listwise mode). Qwen3-8B-Q4 reranks BGE-M3 audited-rebuild top-20 candidates per query via 4-document sliding-window setwise prompt (50% overlap, ≤ 16 token output). "vs. Cosine" is relative gain over BGE-M3 cosine baseline. "vs. Ours-GBM-11" is relative gain over our pipeline's deployment-rule output (Table 3). Setwise reranking with a generic open-source LLM *degrades* BGE-M3 cosine baseline on all five corpora (mean −29.5%); our geometric pipeline (Ours-GBM-11) exceeds setwise reranking on every corpus, with the largest absolute margin on FiQA (+0.171 NDCG@10).

Dataset	Cosine (BGE-M3)	Setwise (Qwen3-8B-Q4)	vs. Cosine	Ours (GBM-11)	vs. Ours
NFCorpus	0.2895	0.2532	−12.5%	**0.3297**	−23.2%
SciFact	0.6406	0.3825	−40.3%	**0.5418**	−29.4%
ArguAna	0.3298	0.2426	−26.4%	**0.4862**	−50.1%
SCIDOCS	0.1676	0.1506	−10.1%	**0.2451**	−38.6%
FiQA	0.4038	0.2616	−35.2%	**0.4331**	−39.6%
Macro	0.3663	0.2581	−29.5%	*0.4072*	−36.6%

Table 16: Falsification record (21 approaches). Detailed analysis of items 1, 7, 20, 21 above; remainder summarized here.

#	Approach	Outcome	One-line root cause
1	Convection-diffusion advection	5/5 harmful	Curse of orthogonality in $\mathbb{R}^{4096}$
2	Allen-Cahn reaction term	Dormant	$O(\Delta^3)$: short-query score range too narrow
3	Allen-Cahn + normalization	-12.9%	Rescaling $\Rightarrow \gamma_{\text{eff}} = 500$, divergence
4	JL random projection (4096→128)	-7.6%	Information loss > noise reduction
5	"Flow-field annealing"	Falsified	JL variance amplification is random noise
6	Scalar potential (Darcy)	NDCG↓	Collapses to isotropic label propagation
7	Density-weighted Chamfer	-1.3%	Dense = redundancy, not importance
8	PQ code reconstruction	-34%	min amplifies quantization error
9	PQ-Chamfer initial retrieval	-4.2%	Chamfer degenerates for single-point query
10	Global graph diffusion	-7.6%	BFS noise dilution > graph benefit
11	L3 subspace cross-visit	No gain	LID: Qwen3 dims uniformly distributed
12	Binary HDC + popcount	N/A	Bottleneck is not distance computation
13	Turing patterns	Impossible	Requires two coupled species
14	Tensor diffusion	Infeasible	4096^2 matrix per edge $\approx$ 6 TB
15	Mixed initial field	Inconsistent	Domain-dependent: $+3.2\%$ / -1.5%
16	Temperature-scaled advection	Divergent	Pe=0.604 exceeds stability limit
17	Multi-dimensional alignment	Unstable	0% convergence, 85.8 ms
18	Block-aligned attention	Unstable	20% convergence, Pe=0.226
19	PCA local projection	No net gain	33 ms latency, marginal accuracy
20	Token on strong baseline	-3%	Cosine 0.47 already saturated
21	Multi-emission coarse filter	NDCG unchanged	Recall+22% but marginal documents only

Table 17: Cross-model generalization on NFCorpus (323 queries).

Model	Parameters	Dim	Cosine	+Graph Sm.	Rel. Gain	Vectors
Qwen3-Embed-8B	8B (Q4)	4096	0.2195	0.2900	**+32.1%**	original
BGE-M3	568M	1024	0.3113	0.3067	-1.50%	audited rebuild[‡]
BGE-large-en-v1.5	335M	1024	0.2975	0.3059	**+2.8%**	original

[‡] BGE-M3 vectors were re-encoded on 2026-05-01 to 2026-05-02 via `llama-server` serving `bge-m3-f16.gguf`; the original BGE-M3 vectors (which had reported BGE-M3 NFCorpus cosine 0.2591 → +Graph Sm. 0.2751 at +6.2%) were lost due to `.gitignore` excluding vector `.jsonl` files from version control and tarball backups (see Section 4.13 *Vector-provenance note*). Qwen3-Embed-8B and BGE-large-en-v1.5 vectors were preserved in their original encoding. Qwen3-Embed-8B and BGE-large achieve $p < 0.05$ (paired bootstrap, 10,000 iterations).

Table 18: Graph Laplacian smoothing on ColBERTv2 MaxSim baseline (5 BEIR corpora, $k = 10$, $\alpha = 0.15$, $T = 5$, paired bootstrap 10,000 iterations). NFCorpus / SciFact / ArguAna / SCIDOCS measured on RTX 5060 Laptop GPU (CUDA 13.1); FiQA measured on EPYC 7B13 CPU-only (NUMA `-interleave=all`). Per-cell raw outputs: `benchmark/colbertv2_lap_{nfcorpus,scifact,arguana,scidocs,fiqa}_results.json`.

Corpus	ColBERTv2 only	+ Lap	Δ rel	p	verdict
NFCorpus	0.2754	**0.2853**	+3.56%	0.0001	POSITIVE-SIG
SciFact	0.5989	0.5970	−0.33%	0.6646	noise
ArguAna	0.3195	**0.3403**	+6.51%	0.0001	POSITIVE-SIG (max)
SCIDOCS	0.1549	0.1572	+1.51%	0.0852	POSITIVE-borderline
FiQA	0.2992	0.2979	−0.43%	0.5240	noise

Table 19: Computational cost comparison.

Resource	Trained Models	Ours
Training data	100M+ pairs (BGE) / 500K (ColBERTv2)	None
Training hardware	16–32× A100 (BGE)	None
Inference	GPU recommended	CPU (50 ms/query)
Total cost	$1,000–5,000+	**$47**